HEART: Hybrid Evidence-based AI Risk Taxonomy for General, Agentic, and Physical AI

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Abstract

AI is becoming more capable, adaptive, and embedded in everyday life. AI systems increasingly combine general-purpose models, memory-equipped agents, and physical actuation, so that potential risks can arise across interpretation, delegated decision-making, and physical contact with people. This integration raises two complementary problems. The mechanistic problem is that domain-specific classifications can identify individual risks without making their interconnections explicit. The methodological problem is that a taxonomy must incorporate emerging evidence while keeping its categories consistent: fixed frameworks may overlook new mechanisms, whereas unconstrained, evidence-driven grouping may yield overlapping categories and uneven granularity. This report presents HEART, a Hybrid Evidence-based AI Risk Taxonomy, which addresses the methodological problem through hybrid construction and supports service impact assessment through service-specific potential-risk profiling. Top-down concepts drawn from prior research, governance frameworks, and safety standards are reconciled with bottom-up evidence through human expert review. The resulting hierarchy organizes 622 bilingual, evidence-linked risk cards into 55 risk categories across General AI (general-purpose AI models), Agentic AI, and Physical AI; each card retains a stable primary classification and documented evidence lineage. Application profiles use the common hierarchy to identify potential risks relevant to a target service and to record why each candidate was included or excluded. An urban delivery-robot service illustrates how the same taxonomy can support potential-risk discovery across different service contexts. The detailed delivery-robot profile spans order intake, route planning, sidewalk navigation, recipient handoff, remote assistance, and maintenance. These profiles produce reviewable candidate sets; they do not establish causal relationships, incident probabilities, severity, or the effectiveness of any mitigation. HEART thus contributes a traceable method for taxonomy construction and maintenance, together with a common vocabulary for discovering and comparing potential risks before service-specific evaluation and mitigation decisions.

Keywords AI risk taxonomy; evidence-based taxonomy; General AI; Agentic AI; Physical AI; risk cards; human-in-the-loop governance.

1 Introduction

1.1 Why an AI Risk Taxonomy Is Needed Now

As AI becomes more capable and adaptive, general-purpose models increasingly support systems that remember prior interactions, plan through tools, coordinate with other agents, and operate in shared physical environments. Established concerns such as misinformation, privacy loss, discrimination, and security failure therefore remain relevant, but their consequences depend increasingly on how interpretation, delegated authority, and execution are connected. For example, a misunderstood instruction can become an agent's action plan and then a hazardous robot handover, whereas effective clarification, permission checks, or independent safety controls can interrupt that pathway. Identifying each constituent risk is thus necessary but not sufficient for assessing an integrated service.

The first problem is mechanistic, because a single-domain hierarchy does not specify how separate failures connect in a deployment even though sociotechnical and embodied-AI research examines broader harms and pipeline-level failures [1–3]. For example, long-horizon robotic safety research remains fragmented across planning, control policy, and execution, even though semantic misgrounding, subtask errors, and execution

drift can accumulate within one system [4]. A taxonomy must therefore distinguish inherited model risks from agency- and embodiment-specific mechanisms while supporting service-specific analysis of their interactions.

The second problem is methodological, since fixed upper structures may fit unfamiliar evidence poorly, whereas unconstrained grouping may reproduce corpus biases, duplicate mechanisms, and yield uneven granularity. Although iterative taxonomy development combines conceptual and empirical approaches [5, 6], that balance must here accommodate changing capabilities, bilingual risk cards (Section 3), and cross-domain boundaries while preserving comparability.

HEART (Hybrid Evidence-based AI Risk Taxonomy) addresses these problems through human-expert-led hybrid construction and service-specific application profiles. While hybrid construction reconciles prior concepts with emerging evidence, application profiles use the resulting hierarchy to identify potential risks across inherited domains and document their relevance to service stages, affected parties, assets, and operating conditions.

1.2 General AI, Agentic AI, and Physical AI Terminology

HEART organizes AI risks under three **operating domains**. Because these labels are used inconsistently in the literature, this report defines them explicitly before use.

Operating-domain terms

General AI denotes general-purpose AI models that display significant generality, competently perform a wide range of distinct tasks, and can be integrated into diverse downstream systems [7, 8]; in this report, their primary inputs and outputs are digital content such as text, images, audio, video, and code. The label does *not* refer to artificial general intelligence (AGI).

Agentic AI denotes AI systems that pursue goals with limited human supervision by planning and taking consequential actions, typically through tool use, persistent memory or state, delegation, or interaction with other agents [9, 10]. They act on external systems directly rather than only producing content for a person to act on.

Physical AI denotes AI systems, such as robots, autonomous vehicles, and humanoids, that perceive, decide, and act in the physical world through a material perception–action loop in which sensing, world-state estimation, control, actuation, mechanical integrity, or physical human–system interaction is constitutive [2, 3].

Because an agent usually relies on a general-purpose model and a robot may incorporate both, the domains are nested in practice. HEART assigns a risk to Agentic AI or Physical AI only when agency or embodiment is constitutive of its mechanism; otherwise, services inherit its General AI classification (Section 3.3). Hereafter, the capitalized domain names denote L1 categories, while "general-purpose model" denotes the underlying technology.

1.3 Research Questions and Design Requirements

The study addresses two related research questions. Whereas **RQ1 (mechanistic)** asks how a common classification can preserve distinct mechanisms while supporting potential-risk discovery across General AI, Agentic AI, and Physical AI services, **RQ2 (methodological)** asks how emerging evidence can be incorporated without uncontrolled duplication or unstable boundaries. Both questions are motivated by KT service impact assessments, in which model disclosure, agent-mediated transmission, and robotic recording of sensitive data require different evidence and context checks despite sharing a broad harm label.

These questions impose four design requirements, which Section 3.6 operationalizes as six design principles. Risks must be specific enough to be identified and reviewed as potential service risks, while each card must retain a stable primary path and traceable evidence. At the same time, computational suggestions must remain subject to semantic adjudication, and application-specific relationships must preserve category definitions. Complementary foundations for these requirements come from the NIST AI Risk Management Framework (AI RMF), ISO/IEC 23894, machinery risk-assessment principles, and the EU AI Act [7, 11–13].

1.4 Scope of the Report

HEART provides a common basis for identifying potential risks during service impact assessment by organizing them into System, Interaction, and Societal Impact areas across General AI, Agentic AI, and Physical AI. Within this scope, evaluation of likelihood and severity, mitigation selection, responsibility allocation, and residual-risk acceptance remain subsequent service-specific activities.

The analytical scope covers the AI lifecycle, including data, model behavior, evaluation, tools, delegated decisions, sensing, control, human interaction, and institutional deployment. To illustrate its application, a hypothetical urban delivery-robot service (Section 6) profiles inherited risks across six service stages, from order intake to maintenance, providing an illustrative application profile rather than a field study.

1.5 Contributions

The report makes four contributions, which address the research questions through taxonomy construction and application.

1. A hybrid construction and maintenance procedure reconciles prior concepts with heterogeneous evidence through explicit boundary review, consolidation, and human expert adjudication (RQ2).
2. A five-level hierarchy contains 55 formal categories and 622 bilingual risk cards, preserving primary classifications and evidence links across the three operating domains (RQ2).
3. Service-specific mapping distinguishes stable classification from context-dependent applicability, allowing common categories to support potential-risk discovery across services (RQ1).
4. Service profiles for an urban delivery robot, a Financial AI Agent, and a Medical AI Agent illustrate reuse across different contexts; the delivery-robot profile provides the detailed worked example (RQ1).

The claimed value lies in traceable organization and analytical reuse; effects on deployment safety and comparisons with alternative construction methods are left to future work (Section 7).

Section 2 reviews prior work, Section 3 defines the design rationale, and Section 4 describes construction. Section 5 reports the hierarchy and algorithm evaluations, followed by the service example in Section 6, limitations in Section 7, and conclusions in Section 8.

2 Prior Methodological Basis

2.1 Risk Repositories, Frameworks, and Taxonomy Development

Risk repositories and survey taxonomies provide broad coverage and a vocabulary for comparing harms, as illustrated by the MIT AI Risk Repository's synthesis of multiple classification systems into a causal and domain-oriented resource and by language-model taxonomies covering discrimination, misinformation, malicious use, human-computer interaction, and broader social effects [14, 15]. These organizing choices also show that category boundaries depend on analytical purpose, so a repository optimized for comparative retrieval does not necessarily provide the atomic, stable, and traceable units required for system-level assessment.

Taxonomy-development research provides a complementary methodological basis by formalizing iterative conceptual-to-empirical and empirical-to-conceptual development around a defined meta-characteristic and explicit ending conditions [5], while subsequent guidance emphasizes transparent construction decisions and explicit evaluation [6]. Drawing on this methodological basis, the present study derives upper-level concepts from prior frameworks, examines evidence inductively, and subjects computational proposals to documented human expert review rather than relying on clustering alone. Table 1 summarizes the principal organizing structures of the prior taxonomies, governance frameworks, and standards discussed in this section.

A meta-characteristic is the overarching perspective that guides which distinctions a taxonomy should represent; in HEART's recorded category decisions, this perspective prioritizes the defining risk mechanism and its semantic boundary against neighboring categories. This account describes the organizing perspective evident in the decisions without implying that a separate formal meta-characteristic was declared before construction.

Similarly, ending conditions specify when a taxonomy-development cycle is sufficiently complete for its intended release. In HEART, the documented checks comprised review of category boundaries and existing card assignments, unique identifiers, complete required fields, nonempty L3 categories, and consistent parent paths. Although these checks supported release-level acceptance, the records do not establish that a complete stopping checklist was specified in advance or that the resulting categories exhaust the risk space.

Table 1. Prior risk taxonomies, governance frameworks, and standards, with their principal organizing structures.

Work	Primary orientation	Principal organizing structure
Weidinger et al. [15]	Language-model harm	Discrimination, hate speech, and exclusion; information hazards; misinformation harms; malicious uses; human–computer interaction harms; environmental and socioeconomic harms
MIT AI Risk Repository [14]	Meta-review and repository	Causal taxonomy by entity, intent, and timing; domain taxonomy spanning discrimination and toxicity, privacy and security, misinformation, malicious actors and misuse, human–computer interaction, socioeconomic and environmental harms, and AI system safety, failures, and limitations
NIST AI RMF [11]	Lifecycle governance	Govern, Map, Measure, and Manage functions
NIST Generative AI Profile [8]	General-purpose generative AI	Cross-sectoral risks and actions organized as a profile of the AI RMF functions
OECD classification framework [16]	Policy-oriented system characterization	People and planet; economic context; data and input; AI model; task and output
OWASP Top 10 for LLM Applications [17]	Application security	LLM application vulnerabilities, including prompt injection, sensitive-information disclosure, excessive agency, supply-chain risk, and unbounded consumption
MITRE ATLAS [18]	Adversarial threat knowledge	Tactics, techniques, procedures, mitigations, and case studies for attacks on AI-enabled systems
ISO 12100 [13]	Machinery safety	Hazard identification, risk estimation, risk reduction, and protective measures

Alongside taxonomy-development methods, governance and safety standards inform impact assessment, risk management, and functional-safety considerations [12, 19, 20], while incident-reporting frameworks distinguish realized harm from hazardous events or conditions that could lead to harm, despite incomplete incident reporting [21]. The attack-focused resources in Table 1 further inform security review, although accidental failures and deliberate attacks remain analytically distinct.

2.2 General AI and Sociotechnical Harm

General-purpose and generative models require categories that travel across applications without losing deployment context. The NIST Generative AI Profile connects model-level concerns to lifecycle risk-management actions, while the OECD classification framework characterizes systems through people and planet, economic context, data and input, model, and task and output dimensions [8, 16]. Sociotechnical harm research further distinguishes representational, allocative, quality-of-service, interpersonal, and societal harms, showing that model behavior and institutional use cannot be assessed in isolation [1]. These works support a broad General AI domain, but they do not by themselves supply the stable, evidence-linked risk cards and **inheritance rules** needed when the same model becomes part of an agent or embodied system. These rules ensure that model-level risks remain in scope when a model is incorporated into an agent or embodied system while retaining their original classification, thereby avoiding duplicate categories and supporting consistent assessment across services.

2.3 Agentic AI Governance and Evaluation

Agentic systems change the unit of analysis from an isolated output to a sequence of stateful decisions and actions, bringing harms associated with delegated goals, authority, persistence, monitoring, and accountability into focus [9, 10]. Evaluation studies complement this governance perspective by exposing operational failure modes, with AgentBench testing reasoning and decision-making across interactive environments and AgentDojo evaluating utility and prompt-injection security when agents act through tools over untrusted data [22, 23]. Together, these resources justify dedicated Agentic AI categories for goal pursuit, excessive authority, traceability, coordination, and cascading behavior while preserving inherited General AI risks.

2.4 Embodied and Cross-Domain Risk Taxonomies

Policy-derived and embodied-AI taxonomies clarify why an upper hierarchy cannot be designed from text-safety categories alone. AIR 2024 derived 314 categories from government and corporate policies and organized them under System and Operational, Content Safety, Societal, and Legal and Rights risks [24]. This separation supports an upper structure that distinguishes where a failure is generated and governed from the type of harm eventually observed.

Recent embodied-AI research adds complementary technical and policy views. Li et al. trace safety issues across perception, cognition, planning, action, interaction, and agentic coordination, including endogenous failures and adversarial vulnerabilities that can propagate through an embodied pipeline [2]. Perlo et al. organize embodied-AI risks into physical, informational, economic, and social dimensions and identify policy gaps that extend beyond conventional robot-safety rules [3]. Safe-BeAl shows that non-adversarial task planning can create physical hazards across multiple hazard classes, making hazard-oriented benchmarks useful evidence for Physical AI even though they do not constitute a complete governance taxonomy [25]. Together, these studies motivate a hierarchy that combines operating domain, governable risk location, stable risk family, and evidence-level risk cards rather than adopting any single pipeline, attack, hazard, or impact axis as the whole taxonomy.

Long-horizon manipulation research further links semantic misgrounding, subtask error propagation, and contact-rich execution [4], motivating examination of connected interpretation, planning, and execution mechanisms rather than isolated robot actions. The delivery-robot example in Section 6 draws on this perspective to select categories across these stages.

Complementing these pipeline-level perspectives, human–robot interaction research distinguishes task failures from their effects on human trust and perceptions of the robot [26, 27], thereby informing the human-facing risk mechanisms represented in the taxonomy.

2.5 Mechanistic and Methodological Gaps

Although existing repositories, agent evaluations, embodied pipelines, and human–robot interaction studies provide complementary views, they do not specify how interpretation, permissions, intervention, and action connect in a deployment. HEART therefore supports application-level mapping of these relationships through Sections 3.5 and 6.

Maintaining classification as evidence changes requires balancing top-down interpretability against bottom-up sensitivity to unfamiliar mechanisms. Although iterative methods already address this tension [5, 6], bilingual, cross-domain cards require operational inheritance rules and documented merges, splits, reassignments, and wording changes.

2.6 Hybrid Construction and Context-Specific Application

HEART uses human-expert-led best-fit framework synthesis to code evidence against an a priori framework and revise it where fit is inadequate [28]. In this process, top-down research, governance frameworks, and standards inform the hierarchy and inclusion criteria, while publications, reports, and patents supply bottom-up candidates; human experts then reconcile both directions (Figure 1) and authorize boundary changes when evidence supports distinct mechanisms. “Evidence-based” thus denotes traceable grounding and lineage rather

than automatic induction from frequency. By combining this grounding with cross-domain inheritance, bilingual evidence-linked cards, stable identifiers, and documented maintenance, the framework allows application profiles to record service-specific inclusion or exclusion without treating semantic proximity as causation.

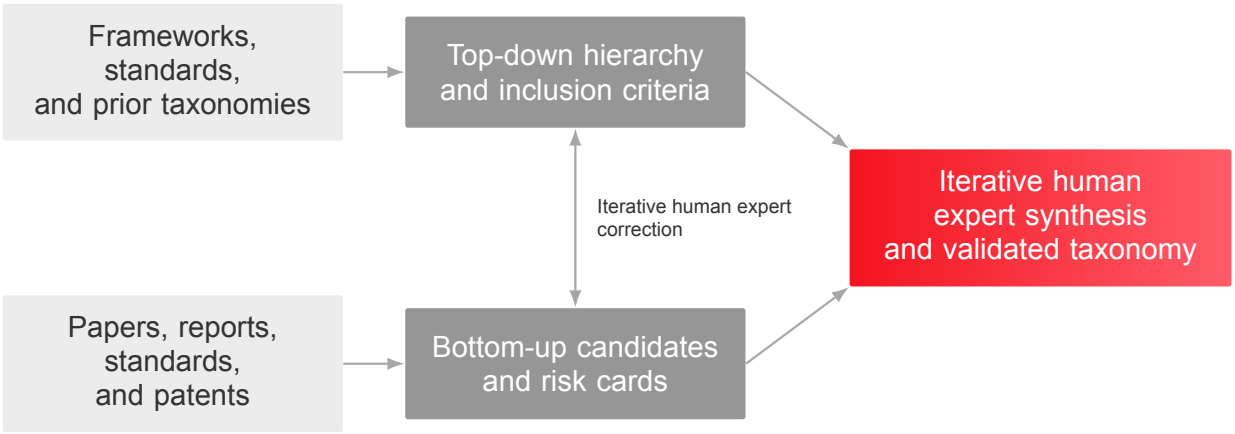


Figure 1. Hybrid response to the methodological gap. Top-down architecture and bottom-up evidence are joined through human expert best-fit synthesis, not through automatic category creation.

3 Scope and Design Rationale

3.1 Scope and Unit of Analysis

Core concepts

Harm is an adverse effect on a person, group, organization, society, environment, asset, or protected interest. A **hazard** is a source or condition with the potential to cause harm. **Risk** is the possibility, with uncertain likelihood and consequence, that an AI-related mechanism, event, or condition produces harm in a stated context. An **incident** is an observed event, or series of events, that causes harm or operational disruption; a **near miss** is an observed event that could plausibly have caused harm but did not. These working definitions are adapted from risk-management and incident-reporting practice [11, 12, 21].

An **AI risk card** is the smallest evidence-linked unit of classification and review. It is a stable, bilingual formulation that identifies the causal mechanism or failure mode, the affected object or protected interest, and the plausible adverse consequence, and it records its primary hierarchy path and supporting evidence. Unlike a keyword, incident narrative, broad outcome, or mitigation, a card is specific enough to support potential-risk identification and review yet general enough to recur across comparable services.

HEART treats the risk card, rather than the source document or observed incident, as its unit of taxonomy construction, covering mechanisms originating in model behavior, data and infrastructure, human interaction, delegated action, multi-agent coordination, sensing, control, physical interaction, and institutional deployment. Hazards in which AI plays no material role are therefore excluded, whereas downstream impacts remain eligible when AI materially contributes to the harm. Even in such cases, speculative downstream effects do not override a card's defining mechanism and approved category boundaries.

3.2 Five-Level Taxonomy Architecture

Table 2. Five-level architecture of the HEART taxonomy.

Level	Taxonomy role	Definition
L0	Common root	Responsible AI (`L0_RAI`) as the common root for organizing potential AI risks.
L1	Operating domain	General AI, Agentic AI, or Physical AI, identified through the risk mechanism and approved category boundaries.
L2	Risk area	System, Interaction, or Societal Impact, identifying the principal locus of the risk mechanism.
L3	Risk category	A stable risk family used to organize, compare, and report potential risks.
L4	Risk card	The atomic bilingual, evidence-linked risk formulation used to identify potential service risks.

Table 2 defines the taxonomy role of each level, and Figure 2 summarizes the HEART construction logic. The upper hierarchy is established top-down from prior taxonomies, governance frameworks, and standards, then refined through human expert review, while evidence is synthesized bottom-up into risk cards. Iterative human expert reconciliation pursues near-MECE boundaries (Section 3.4), resolves duplicate or cross-domain candidates, and permits an upper-level change only through explicit human expert review.

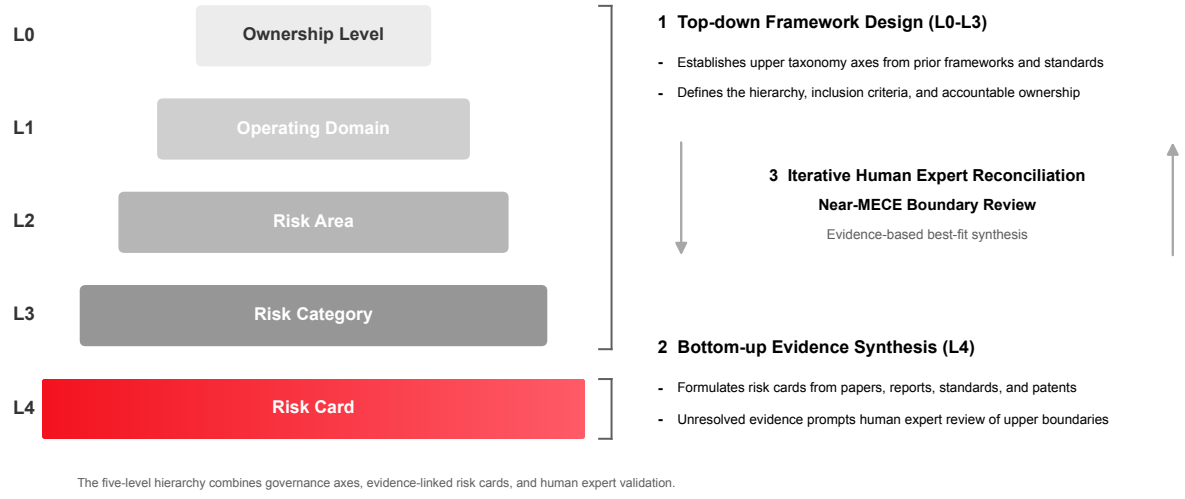


Figure 2. Design principles for the five-level taxonomy. Top-down framework design defines L0–L3; bottom-up evidence synthesis constructs and maps L4 risk cards; iterative human expert reconciliation maintains near-MECE boundaries and controlled hierarchy maintenance.

3.3 Operating Domains and Cross-Domain Inheritance

Section 1.2 defines the three operating domains; for classification, their boundaries are operationalized as follows. **General AI** covers model-level and associated interaction or societal-impact mechanisms that do not require delegated agency or physical embodiment. **Agentic AI** adds autonomous goal pursuit, consequential action, tool use, persistent state, delegation, or multi-agent interaction; the conceptual treatment is consistent with recent agent-governance work [9, 10]. **Physical AI** requires a material perception–action loop in which sensing, world-state estimation, control, actuation, mechanical integrity, or physical human–system interaction is constitutive [2].

Under this inheritance model, an agent built on a general-purpose model can retain misinformation, privacy, and adversarial-security risks, while a robot with an agentic planner can inherit those risks and add excessive authority, coordination, sensor-fusion, control, and human–robot safety risks. The application profile in Section 6 applies this model while preserving each selected category's identifier.

Because inheritance makes the same category potentially applicable to many services, application profiling separates stable taxonomy categories from context-dependent applicability. The discovery stage considers the target service, affected parties, assets, operating environment, data and inputs, model properties, tasks, outputs, autonomy, and physical capabilities. This distinction is consistent with policy-oriented classification approaches that characterize an AI system through its social and economic context, data and inputs, model properties, and task and output [16].

3.4 Near-MECE Design Objective

Because AI failures can span perception, planning, and physical action, HEART pursues near-MECE (mutually exclusive and collectively exhaustive) boundaries by assigning each card one primary category according to its defining mechanism, supported direct consequence, and approved boundaries. This avoids double counting without determining legal, moral, or institutional accountability. Within this approach, incident classification uses documented harm, whereas prospective assessment identifies plausible mechanisms and pathways without establishing occurrence, causation, likelihood, or severity. A possible downstream collision therefore does not determine the classification of an earlier failure, just as a prospective candidate is not presented as an observed incident. Evidence status thus remains separate from hierarchy membership, allowing several connected mechanisms to occur within one scenario while each card retains its own primary path.

Placement follows the L3 inclusion and exclusion criteria in Appendix A. For Physical AI, semantic misidentification belongs to Input Interpretation Error, whereas errors in location, pose, motion, contact, force, or sensor fusion belong to Physical State Estimation and Sensor-Fusion Failure. When an otherwise valid task is translated into unsafe path or motion planning, control, or actuation, the failure belongs to Unsafe Physical Control and Actuation. By comparison, once a person is correctly recognized, failures of separation, slowing, stopping, contact-force limitation, or safe handover belong to Human-Robot Physical Safety Failure.

The same boundary-based approach distinguishes Physical Tampering and Sabotage, which requires physical contact or manipulation of components, including blocking, attaching, replacing, or damaging them, from non-contact signal injection, which remains an adversarial-security risk. For an integrated vision-language-action policy, placement follows the observed failure symptom rather than an inferred internal module, while API- or IoT-mediated device operation remains Agentic when physical perception, control, and actuation are not the dominant failure mechanism. These rules apply across the 55 categories.

3.5 Classification and Failure-Mode Mapping

A *failure mode* describes how a function, interaction, or control can fail within a stated service, whereas a taxonomy category may represent that mechanism across many applications and a service pathway links several mechanisms under explicit conditions. However, not every risk is a functional failure, since harmful outcomes may also arise from intended behavior, institutional choices, or misuse. The urban delivery-robot profile in Section 6 therefore selects categories relevant to service stages without establishing the conditions connecting them, which would require further failure-mode analysis and supporting evidence.

A proposed pathway should identify a triggering condition, the relevant category IDs, the condition permitting continuation, the plausible consequence, and possible intervention points. Because these links are analytical hypotheses, they must be supported or updated with service-specific evidence rather than inferred from shared domain membership or embedding similarity. Several categories can consequently describe one scenario without duplicating a card's primary path, while recording pathways separately from the hierarchy distinguishes stable classification, which serves RQ2, from the analysis of combined behavior, which serves RQ1.

3.6 Design and Update Principles

Six principles operationalize the requirements in Section 1.3. Atomicity and duplicate control support specific, nonredundant candidates; source traceability and stable identifiers preserve evidence-linked primary paths; human expert approval governs computational suggestions; and definitional consistency enables reuse across applications. Updates must preserve these relationships while allowing newly evidenced mechanisms and documented boundary changes. HEART supports risk discovery and organization without claiming exhaustive coverage, certification, or allocation of responsibility.

4 Methodology

4.1 Construction Procedure

HEART was constructed through deductive, inductive, and adjudicative stages, with deductive design establishing provisional upper-level concepts and inclusion criteria, inductive synthesis retrieving and consolidating candidate mechanisms from heterogeneous evidence, and adjudication assessing fit, duplication, and boundary implications. Together, these stages produced a bilingual taxonomy that assigns each card one primary hierarchy path and records its derivation. Because source documents provide evidence, candidate statements express proposed formulations, and cards constitute the smallest classified units, the statistical unit changes across stages; their counts therefore represent neither a one-to-one conversion rate nor an estimate of risk prevalence.

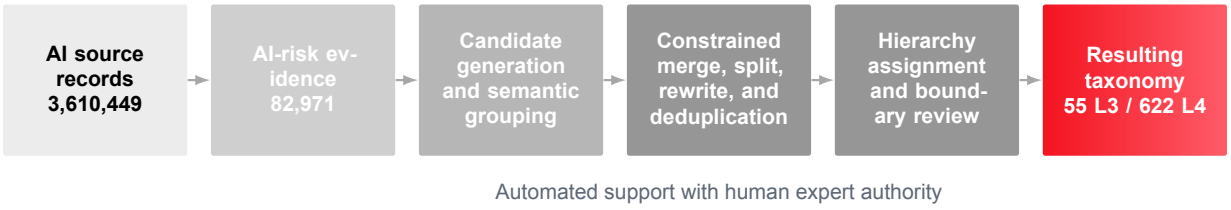


Figure 3. Taxonomy construction procedure. Evidence records were synthesized and adjudicated into risk cards; the stages use distinct statistical units.

Top-down design supplies the hierarchy and inclusion criteria, after which corpus construction and screening, candidate generation and consolidation, and hierarchy-assignment proposals proceed under human expert review (Figure 3). Automated grouping and assignment support this review without authorizing category changes, while the resulting structure undergoes the verification reported in Section 5. In this way, prior concepts constrain abstraction and inclusion, evidence can reveal missing distinctions, and human expert adjudication reconciles the two, providing the rationale for the hybrid design.

4.2 Evidence Space and Corpus Construction

4.2.1 Evidence Sources

The evidence space combines scientific publications, policy and technical reports, and patents to capture complementary forms of evidence. While scientific literature contributes observed failures, evaluation results, and emerging mechanisms, policy and technical reports add institutional concerns and governance requirements, and patents indicate intended capabilities and deployment mechanisms before incidents are widely reported.

Each record retains, when available, a source identifier, title, abstract or executive summary, publication year, authors or issuing organization, DOI or URL, source family, retrieval date, and processing provenance. Because source diversity is itself informative, independent sources are not collapsed merely because they describe the same risk; instead, exact duplicates within a source type are removed using persistent identifiers first and normalized title keys second. Within these records, processing provenance describes how sources were retrieved and transformed, whereas evidence lineage connects them to candidate formulations, consolidation decisions, and final risk cards so that retained links can be followed during review.

4.2.2 Corpus Integration and Screening

The construction baseline contains 3,610,449 normalized AI source records from the 1950s through July 2026, from which risk-focused source rules, lexical markers, semantic retrieval, and human review retained 82,971 AI-risk evidence records. The retained records describe an AI failure, hazard, misuse, vulnerability, unsafe interaction, governance gap, or adverse social effect, and Table 3 summarizes this funnel alongside the statistical unit at each stage.

Table 3. Evidence-to-taxonomy construction baseline.

Stage	Count	Statistical unit
AI source space	3,610,449	Normalized records from heterogeneous AI sources
AI-risk evidence	82,971	Records retained by reviewed source, semantic, and lexical screening
Final taxonomy	622	Bilingual L4 risk cards after consolidation and human expert review

The 82,971 records include 50,375 scholarly publications, 938 policy and report records, and 31,658 patents. Resolved author affiliations of the scholarly and policy/report records span 164 countries, led by China and the United States.

4.2.3 Tri-Domain Filtering

The semantic-retrieval component of this screening relies on domain-specific anchors. For each document d_i, the retrieval text is formed by concatenation,

$$x_i = \text{title}(d_i) \| \text{abstract}(d_i) \| \text{keywords}(d_i). \tag{1}$$

Here $\|$ denotes string concatenation, and reviewed definitions and examples form anchor sets $\mathcal{A}_G$, $\mathcal{A}_A$, and $\mathcal{A}_P$ for General AI, Agentic AI, and Physical AI. As reviewed reference texts describing a domain or risk mechanism, these anchors provide a relevance signal for retrieval without treating similarity as a final classification. A record may therefore be retained when it belongs to a reviewed risk source, is semantically close to a domain or risk anchor, or contains explicit risk, safety, failure, security, misuse, ethics, or governance markers that survive review. Cross-domain candidates likewise remain eligible for review against their stated mechanism, supporting evidence, and approved category boundaries, while prospective risk cards do not require an observed incident.

4.3 Risk Candidate Generation and Consolidation

4.3.1 Semantic Representation and Clustering

Candidate formulations are extracted or drafted from the retained evidence records so that each states the AI role, mechanism, affected object, and plausible adverse consequence (Algorithm 1, step 3); as noted in Section 4, records and candidates are not in one-to-one correspondence. Candidate text combines a provisional name, definition, evidence sentence, and normalized keywords. The BGE-M3 embedding model provides multilingual dense representations, HDBSCAN (hierarchical density-based spatial clustering of applications with noise) or related density-based methods identify candidate neighborhoods, and class-based TF-IDF (term frequency–inverse document frequency) provides interpretable lexical summaries [29–31]. UMAP (uniform manifold approximation and projection) may be used for visualization [32], but final semantic decisions are not made in the UMAP projection.

To illustrate the outcome of this filtering for the narrowest domain, Figure 4 locates Physical AI risk evidence within the broader AI document space.

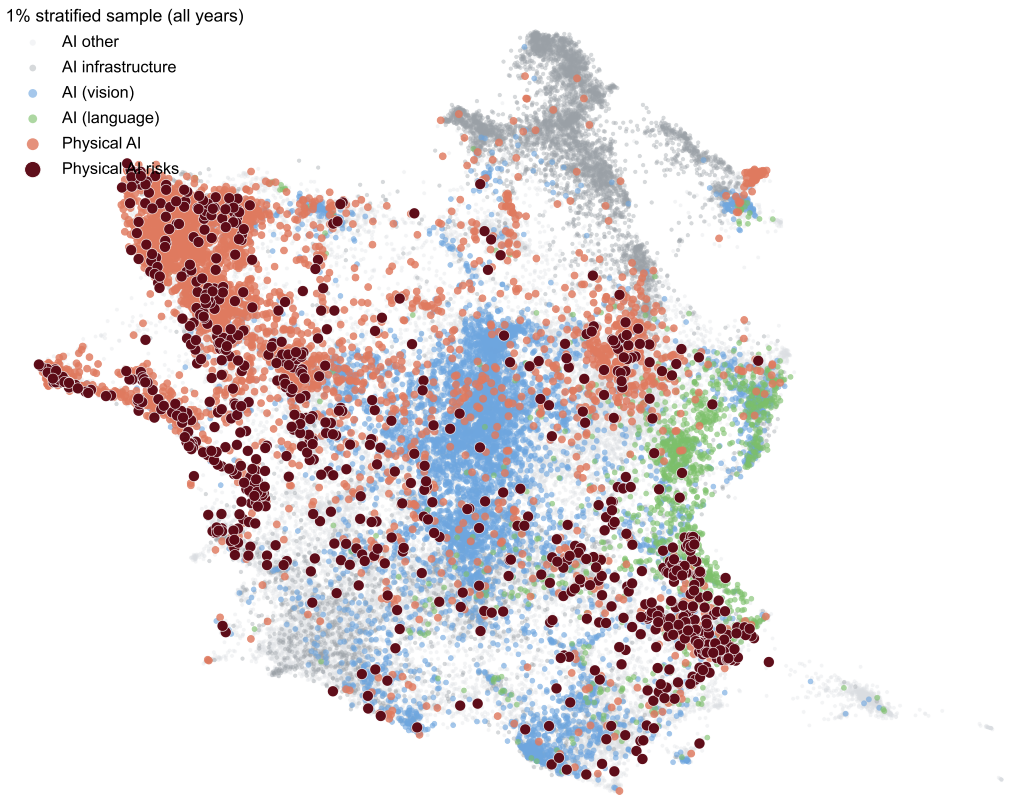


Figure 4. Physical AI document-space diagnostic based on a 1% stratified sample of the integrated AI corpus. Each point is one document, positioned by a two-dimensional projection of its text embedding to summarize semantic neighborhoods; projected distances may distort relationships in the original embedding space, and the axes carry no substantive units and are omitted. Document groups are distinguished by color, with AI other (light gray), AI infrastructure (gray), AI vision (blue), AI language (green), and Physical AI (orange). Physical AI risk evidence retained by screening is overlaid as larger dark-red points. The overlay concentrates within and along the edges of the Physical AI regions but also extends into neighboring regions, which is why cross-domain candidates require review against supporting evidence and approved category boundaries.

An embedding represents a text as a numerical vector, and semantic similarity compares the proximity of such vectors to identify potentially related formulations for review; it does not establish that the formulations are equivalent or causally connected.

If $z_i = \mathrm{Enc}(c_i)$ is the unit-normalized embedding of candidate text c_i, cosine similarity provides the retrieval signal

$$\rho_{ij} = z_i^\top z_j. \tag{2}$$

High similarity triggers duplicate or boundary review; it does not itself authorize a merge.

For the iterative grouping evaluation, a descending similarity threshold connects 1,612 candidates into components, with each iteration selecting the grouping at the first crossing of within-component cohesion and external attraction. Whereas cohesion pools similarities among members of non-singleton components, external attraction averages each component's strongest similarity to representatives outside it. Each component then retains its most central existing candidate, or medoid, for the next iteration, thereby reducing the inventory without rewriting definitions or establishing semantic equivalence.

4.3.2 Constrained Consolidation

Consolidation converts semantically grouped candidates into reviewable risk cards, but similarity only identifies candidates for comparison and does not authorize a merge. Candidates are therefore merged only when their risk mechanisms, affected interests, and consequences are substantively equivalent at a comparable level

of abstraction and require no distinct assessments or controls, whereas differences in protected interest, control owner, causal stage, or evaluation requirements justify retaining separate cards. In turn, composite formulations are split into independently assessable mechanisms, scenario-specific statements are generalized only when supported by evidence, duplicates are removed, and statements without a material AI role are excluded.

Surviving formulations are drafted as cards with stable identifiers, bilingual names and definitions, hierarchy paths, and evidence links, with each definition identifying the affected target, mechanism, and consequence. Human experts then confirm, modify, or reverse the proposed operations, while derivation records retain merge and split links and retired identifiers remain unavailable for reuse. Following hierarchy assignment and boundary review (Section 4.4), this process yields 622 bilingual cards. Algorithm 1 in Appendix B describes the full workflow, whereas the numerical evaluation in Section 5.4 assesses only repeated grouping before human expert rewriting, not the complete consolidation and adjudication process.

4.3.3 Human Expert Review and Adjudication

Human experts refined, merged, and removed algorithmically grouped formulations to produce the 622-card reference, balancing conceptual preservation with practical manageability through review of validity, atomicity, bilingual equivalence, duplication, risk mechanism, and hierarchical fit. L3 definitions were independently designed from the literature and human expert judgment, against which observed cases were assigned according to direct harm and prospective cards were assessed without assuming that an incident had occurred. Where cases remained ambiguous, further review and adjudicated decisions took precedence over automated proposals, while category definitions changed only through explicit adjudication.

Upper-level category validation compares every L4 card with each proposed L3 definition and records inclusion and exclusion reasons, accepting a category only when the inventory supports a distinct failure mechanism and a defensible boundary. Because card counts indicate evidentiary support rather than quotas, three well-fitting cards are preferred, although one or two may be accepted when semantic fit is strong. Accordingly, unrelated cards are never moved merely to populate categories, and validation evaluates existing L4 cards without creating new ones.

The resulting cards and assignments thus provide a human-expert-consensus reference for comparison rather than a uniquely correct taxonomy or a target size the algorithm must reproduce. Although derivation records support card-level traceability, complete pre-adjudication ratings were not retained, leaving inter-rater reliability to future independent coding. The automated grouping and assignment evaluation in Section 5.4 is therefore reported separately from human expert adjudication.

4.4 Hierarchical Assignment to L3 Families

4.4.1 Representation and Objective

Semantic assignment represented each L4 card and L3 definition in the same bilingual embedding space. In this subsection and in Algorithm 2, an L3 category is called a *family*, following clustering usage. Let r_i denote card i, d_k category definition k, μ_k its reviewed centroid, and a_i the family to which card i is assigned. The following objective measures within-family semantic coherence, which the assignment procedure seeks to increase subject to domain eligibility and protected assignments.

$$J^{(t)} = \frac{1}{|\mathcal{R}^{(t)}|} \sum_{i \in \mathcal{R}^{(t)}} \cos\left(\mathrm{Enc}(r_i), \mu^{(t)}_{a_i^{(t)}} \right). \tag{3}$$

Here $\mathcal{R}^{(t)}$ contains the cards assigned to a family at iteration t, including all 622 cards in the reported experiment, so the objective serves as a geometric diagnostic rather than evidence that the assigned family is semantically correct.

4.4.2 EM-Style Constrained Assignment and Centroid Refinement

The assignment procedure alternates category proposals with centroid refinement, borrowing the iterative structure of Expectation–Maximization (EM) [33]. Because cards receive hard assignments and the objective mea-

sures semantic coherence rather than likelihood, the procedure is EM-style rather than probabilistic EM.

Algorithm 2 in Appendix B operationalizes this idea. Let $z_i = \text{Enc}(r_i)/\|\text{Enc}(r_i)\|$ denote the unit-normalized embedding of card i ($i = 1, \ldots, N$) and s_k the unit-normalized embedding of the bilingual seed definition d_k of L3 family k ($k = 1, \ldots, K$). Let $\mathcal{E}_k$ be the set of reviewer-approved exemplar cards for family k, $\mathcal{K}_i \subseteq \{1, \ldots, K\}$ the families eligible for card i under the domain constraints, and $\mathcal{P}$ the set of protected cards whose assignments reviewers have already fixed. With $\mathcal{N}(v) = v/\|v\|$ denoting vector normalization, the procedure runs as follows.

1. **Initialization.** Each centroid starts from its seed definition and approved exemplars.

$$\mu_k^{(0)} = \mathcal{N}\Big(s_k + \sum_{j \in \mathcal{E}_k} z_j\Big). \tag{4}$$

   Starting from reviewed definitions rather than from random points anchors each centroid to the intended meaning of its family.

2. **Assignment step (E-step analog).** At iteration t, every unprotected card $i \notin \mathcal{P}$ ranks its eligible families $k \in \mathcal{K}_i$ by the cosine similarity $\cos(z_i, \mu_k^{(t-1)}) = z_i^\top \mu_k^{(t-1)}$. The procedure records for review the ranked candidate families and the top-one margin, that is, the similarity gap between the best and second-best families.

$$m_i^{(t)} = \cos\big(z_i, \mu_{(1)}^{(t-1)}\big) - \cos\big(z_i, \mu_{(2)}^{(t-1)}\big). \tag{5}$$

   The margin is defined only when at least two eligible families are available.

3. **Constraint step.** Candidate families that would violate the direct-harm placement rule, domain eligibility, an explicit reviewer decision, or an approved L3 boundary are removed. Each unprotected card is then provisionally assigned to the highest-ranked remaining family,

$$a_i^{(t)} = \arg\max_{k \in \mathcal{K}_i^*} \cos\big(z_i, \mu_k^{(t-1)}\big), \tag{6}$$

   where $\mathcal{K}_i^* \subseteq \mathcal{K}_i$ is the constrained candidate set. If this set is empty, the card remains pending human expert adjudication rather than being assigned automatically. Pending cards are excluded from centroid updates and the assignment objective until resolved. Protected cards keep their reviewed assignments.

4. **Centroid-update step (M-step analog).** Each centroid is recomputed from its seed and its current members, $\mathcal{A}_k^{(t)} = \mathcal{E}_k \cup \{i : a_i^{(t)} = k\}$, using

$$\mu_k^{(t)} = \mathcal{N}\Big(s_k + \sum_{i \in \mathcal{A}_k^{(t)}} z_i\Big). \tag{7}$$

   Retaining the seed preserves a contribution from the category definition, although its relative influence decreases as membership grows. Normalization keeps the update on the unit sphere, as in spherical k-means [34].

5. **Convergence check.** Iteration stops when no eligible assignment changes or when $|J^{(t)} - J^{(t-1)}| < \epsilon$ with an unchanged assigned-card set; otherwise steps 2–4 repeat until the iteration limit T is reached. If no card can be assigned, automatic iteration stops and unresolved cards proceed to human expert adjudication.

6. **Human expert review and output.** The resulting assignments remain proposals. Small margins, unstable assignments, or conflicts with semantic placement rules or reviewer instructions require human expert review, including cross-domain and Physical AI boundary cases. The procedure outputs proposed families, ranked alternatives, and diagnostics; human experts determine the final classification against reviewed L3 definitions (Section 4.3.3).

Hard assignments resemble classification EM [35], but the seed term and semantic constraints modify the spherical k-means update, so monotonic improvement of J is not guaranteed. Termination uses assignment stability, an objective-change tolerance, and an iteration limit; tolerance-based termination alone does not establish a fixed point.

5 Results and Evaluation

5.1 Resulting Hierarchy and Risk-Card Inventory

The resulting HEART taxonomy contains one L0 common root, three L1 operating domains, seven populated L2 branches, 55 formal L3 categories, and 622 active L4 cards (Table 4). Within this structure, General AI contains 33 categories and 491 cards, Agentic AI 13 categories and 68 cards, and Physical AI 9 categories and 63 cards, with Table 5 breaking the categories down by L2 risk area and Figure 5 showing the card distribution by domain. Every L3 category contains at least one card, while full-inventory semantic adjudication confirmed the 55 category boundaries and all 622 primary assignments. Complementary structural checks confirmed unique identifiers, the absence of exact duplicates, and consistent parent paths.

Table 4. Final hierarchy and active-card distribution.

L1 domain	Populated L2 branches	L3 categories	Active L4 cards
General AI	3	33	491
Agentic AI	2	13	68
Physical AI	2	9	63
Total	7	55	622

Table 5. Final L2 branch counts.

L1 domain	L2 risk area	L3 categories
General AI	Interaction	14
General AI	System	13
General AI	Societal Impact	6
Agentic AI	System	8
Agentic AI	Interaction	5
Physical AI	System	6
Physical AI	Interaction	3

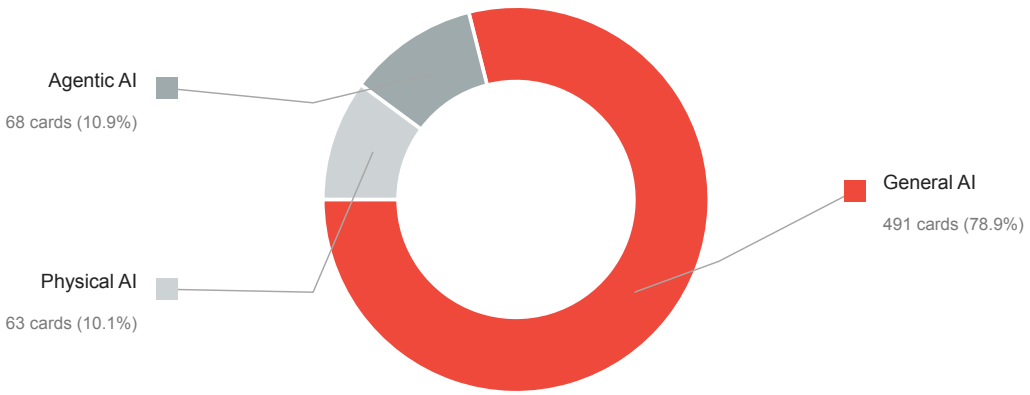


Figure 5. Distribution of risk cards by L1 domain.

The distribution reflects differences in scope, granularity, and available evidence across the three domains, with General AI accounting for the majority of cards (491 of 622) because its categories span content, privacy,

security, evaluation, performance, environmental, economic, civic, and governance mechanisms. Building on this scope, Agentic AI adds distinctive risks from goal pursuit, delegated authority, task execution, persistent state, human oversight, self-correction, deception, traceability, cascading behavior, and inter-agent relations, while Physical AI adds constitutive sensing, state estimation, hazard-aware task planning, timing, safe-state transition, control, actuation, mechanical integrity, human–robot safety, obstruction, and physical tampering. These counts therefore reflect taxonomy granularity and evidence coverage rather than risk prevalence, incident frequency, severity, or likelihood.

5.2 Semantic Space and Cross-Category Structure

The embedding space used for assignment (Section 4.4) also helps identify dense neighborhoods, overlap, outliers, and candidate boundary cases, although its representation is semantic rather than causal. Within this space, General AI categories are expected to span the broadest region because they include interaction and system risks as well as the only Societal Impact branch, whereas Agentic AI categories are expected to concentrate around delegated action, authority, goals, traceability, and coordination. Physical AI categories, in turn, are expected to remain compact because the final boundary excludes robot-context cards whose primary mechanism is general performance, evaluation, security, or governance.

Cross-category proximity is nevertheless expected, since privacy can interact with agent memory and mobile sensing, adversarial security can corrupt perception and produce physical action, and governance failures can influence deployment in every domain. The single-parent hierarchy therefore identifies a primary locus for governance review, while semantic links and application profiles preserve secondary relationships.

5.3 Granularity Criteria and Taxonomy Maintenance

Granularity was reviewed by asking whether mechanisms, affected objects, consequences, and controls are separable, applying the merge and split criteria of Section 4.3.2. At the category level, this means retaining an L3 category when it groups a coherent set of mechanisms rather than acting as an unexplained residual bucket. Accordingly, the taxonomy contains no residual “Others” categories, although newly available evidence may motivate a documented change after duplicate and boundary review. Granularity criteria and semantic similarity thus remain review aids rather than substitutes for the primary-assignment tests of causal proximity, control specificity, definitional fit, and non-duplication (Section 3.4).

5.4 Algorithm Evaluation

The evaluation examines how closely automatic L3 assignment agrees with the human-expert-validated hierarchy and how iterative consolidation groups candidates before human expert rewriting. For this comparison, the final reference contains 55 literature- and advisory-grounded L3 definitions, 622 human-expert-rewritten L4 cards, and their human-expert-confirmed assignments, with the complete 55-category structure validated through comparison of all 622 L4 cards, multi-round human expert review, and final human expert adjudication. The experiments below consequently evaluate automated grouping and assignment separately from that human expert validation process.

The numerical results below use the pre-revision definition text archived with the experiment inputs, whereas Appendix A subsequently integrates classification guidance into standalone definitions and clarifies related category boundaries. Although these editorial revisions preserve the category identifiers and recorded L4 assignments, their effect on embedding-based agreement has not been evaluated.

5.4.1 Common Inputs and Evaluation Scope

BGE-M3 and multilingual-E5-base encode the same ordered English and Korean names and definitions [29, 36], excluding identifiers, hierarchy labels, evidence fields, and human expert assignments from the text. Because seven L3 definitions in the archived experimental inputs contained boundary-routing sentences naming other L3 identifiers, those complete sentences were also excluded to prevent hierarchy-label leakage, while the source definitions used for that experiment were otherwise unchanged. Both models use a 512-token

limit, which no complete input exceeds (maximum lengths of 228 and 231 tokens, respectively), but their tokenization, pooling, and model-specific input formatting differ. The comparison therefore evaluates two representation choices rather than isolating the causal effect of model training.

For Algorithm 2, each card is ranked against categories in its known L1 operating domain, starting from the L3 definition vectors as initial centroids and alternating assignment with definition-anchored centroid updating as described in Section 4.4. No human-expert-assigned exemplars or protected cards are supplied, thereby avoiding direct reuse of the reference answers as fitting constraints. Because the experiment implements domain eligibility and the numerical update without independent machine-readable rules for direct harm, control ownership, or approved semantic boundaries, it evaluates the numerical assignment component rather than the entire human-expert-constrained procedure in Algorithm 2.

Top-1 agreement measures the fraction of cards whose highest-ranked category matches the human expert assignment, whereas Top-k agreement records whether that assignment appears among the first k candidates. Macro F1 complements these measures as the unweighted mean of the per-category F1 scores computed from the Top-1 assignments. Since refinement uses the whole inventory, all three measures describe in-sample reference agreement.

5.4.2 L3 Assignment Results

Table 6. Iterative L3 assignment with complete bilingual text and known L1 domain.

Embedding model	Human-expert-reference agreement			Macro F1
	Top-1	Top-3	Top-5	
BGE-M3	65.11%	84.41%	90.51%	0.5899
Multilingual-E5-base	61.74%	82.32%	88.59%	0.5674

Iterative assignment matches the human expert category for 405 of 622 cards with BGE-M3 and 384 with multilingual-E5-base (Table 6). Direct retrieval against the L3 definitions produces 405 and 389 agreements, respectively. Iteration therefore leaves BGE-M3 Top-1 agreement unchanged and reduces E5 agreement by five cards; macro F1 decreases from 0.5941 to 0.5899 for BGE-M3 and from 0.5759 to 0.5674 for E5. Iterative refinement is consequently not uniformly better than direct retrieval on the completed 55-category hierarchy. The two models propose different categories for 215 cards; these disagreements identify candidates for boundary review rather than indicating which model is correct.

Known operating domains provide informative constraints, as removing them reduces iterative agreement to 61.25% for BGE-M3 and 57.40% for E5. However, ranked alternatives must also be interpreted in relation to the eligible category count. Physical AI, for example, contains nine eligible L3 categories, making its Top-5 agreement interpretable but less selective than Top-1 or Top-3 because it covers more than half of the candidate set. Ranked alternatives should therefore be used to focus human expert review with this candidate-set size in mind.

5.4.3 Input and Model Sensitivity

Table 7. Top-1 human-expert-reference agreement under different text inputs, using iterative L3 assignment with known L1 domains.

Input	BGE-M3	multilingual-E5-base
Bilingual names only	41.32%	40.68%
Korean name and definition	56.27%	54.18%
Bilingual definitions only	51.77%	46.62%
Bilingual input, 128-token limit	62.70%	59.97%
English name and definition	65.27%	59.65%
Complete bilingual name and definition	65.11%	61.74%

Names and definitions provide complementary information (Table 7), since neither field alone reproduces the combined result. Even so, BGE-M3's English-only condition is one agreement higher than its complete bilingual condition, whereas multilingual-E5-base performs best with the complete bilingual representation. Because English appears first, shortening the input also changes which language content is retained, so the truncation condition is not a pure test of text length or translation quality.

Across the predeclared update and stopping conditions, BGE-M3 Top-1 agreement ranges from 63.50% to 66.72%, with 94.37% to 100% assignment stability relative to the primary run. The corresponding E5 ranges are 61.58% to 63.18% and 91.16% to 100%. Under dimension-scaled Gaussian perturbation of 0.10, mean assignment stability is 94.08% for BGE-M3 and 90.77% for E5 across five fixed random seeds. These results show that the main conclusions are directionally stable but that individual assignments and exact agreement values remain sensitive to representation and numerical choices. Termination at a tolerance or iteration limit does not always yield unchanged assignments. Complete settings and machine-readable results are retained with the experiment record.

Algorithm 1 grouping shows a similar dependence on representation. Whereas the construction trajectory yielded 792 representatives after five grouping iterations (F5), re-encoding the available candidate text under matched evaluation conditions yields 817 with BGE-M3 and 875 with E5 at the same stage. These sensitivity runs are not reconstructions of the source encoding, and their cross-model adjusted Rand index of 0.2527 (a value of 1 denotes identical partitions) shows why similar inventory sizes need not imply similar groupings.

5.4.4 Semantic Correspondence to the Human Expert Rewrite

Because human experts used the construction trajectory's F5 output of 792 representatives to rewrite the 622 reference cards, exact wording and one-to-one lineage are inappropriate comparison criteria. Instead, bidirectional semantic retrieval identifies candidate correspondences by defining a reciprocal pair as a representative and reference card that each rank the other first, with Table 8 reporting this measure for F5 and its adjacent stages.

Table 8. Reciprocal nearest-neighbor pairs with the 622-card reference. Counts identify correspondence candidates, not verified semantic equivalence or recall.

Construction stage	Representatives	BGE-M3 evaluator	E5 evaluator
F4	901	502	502
F5	792	498	502
F6	698	451	455

At F5, both evaluators select the same nearest representative for 517 of 622 reference cards (83.12%), indicating agreement between retrieval views rather than independently judged equivalence. Although F6 has fewer reciprocal pairs, these counts depend partly on the size and composition of the representative set, and reciprocal matching cannot capture legitimate many-to-many rewrites. Independently adjudicated pairs would therefore be needed to estimate semantic precision and recall.

Together, the results support automatic grouping and ranked assignment as aids to human expert review, while revealing their dependence on input representation, embedding model, and numerical choices. Human expert adjudication therefore remains the final authority in taxonomy construction, and the choice of a consolidation stage is an operational decision rather than the discovery of an intrinsic optimal scale.

6 Application to Service Impact Assessment

6.1 Context-Specific Taxonomy Composition

This section addresses RQ1 by showing how the common hierarchy supports the identification of potential risks in service impact assessment. An application profile selects relevant L0–L3 paths according to the system's purpose, intended functions, users and affected non-users, operating environment, data flows, autonomy,

physical capabilities, and exception-handling arrangements, while each selected category retains its identifier, definition, and hierarchical relationships. Candidate sets can therefore be compared across services without redefining the taxonomy.

Within these profiles, inherited General AI risks retain their classification rather than being duplicated under Agentic AI or Physical AI. The resulting candidates remain subject to subsequent review, since confirming applicability and evaluating causal pathways, severity, likelihood, and mitigation requires additional service-specific evidence and methods.

The five-step application profile in Figure 6 builds on KT's risk-identification and RAI-assessment approach [37] and adapts it to service-specific use of HEART.

1. define the service boundary and excluded functions;
2. enumerate actors, affected parties, assets, and protected interests;
3. describe data, decision, tool-use, and physical-action flows;
4. select candidate L3 categories from all inherited L1 domains;
5. document applicability, supporting evidence, relevant controls, and accountable parties.

While steps 1–3 characterize the service, step 4 draws candidate categories from every inherited L1 domain so that General AI risks retain their original classification. Step 5 then links selected categories to L4 cards, applicability evidence, relevant controls, and accountable parties in a reviewable discovery record, after which mitigation design can proceed.

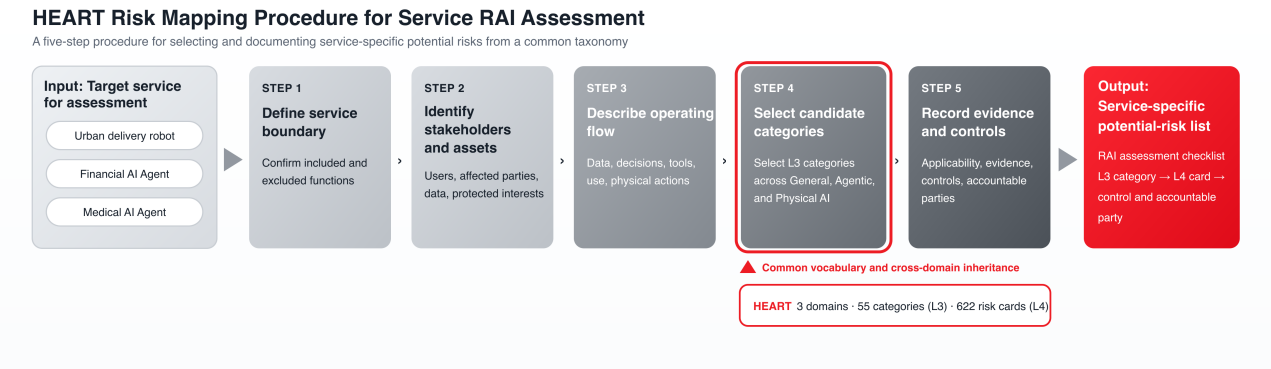


Figure 6. Five-step potential-risk discovery procedure for HEART, informed by KT's risk-identification and RAI-assessment approach. Steps 1–3 characterize the target service, step 4 selects candidate categories from the common HEART hierarchy, and step 5 records applicability, evidence, relevant controls, and accountable parties. The output is a service-specific potential-risk list for subsequent RAI evaluation.

The procedure has supported potential-risk discovery for an urban delivery robot, a Financial AI Agent mapped from 17 sector-specific criteria, and a Medical AI Agent, with each profile reusing HEART identifiers and definitions while selecting service-specific candidates. The delivery-robot case below develops this shared procedure into a detailed worked example.

6.2 Urban Delivery-Robot Example

The delivery-robot service illustrates the profiling procedure through a set of candidate risks for assessment, rather than confirmed risks or an exhaustive account of the risks a delivery robot may present.

The example service is a low-speed, last-mile delivery robot operating on sidewalks and in mixed pedestrian areas. After accepting an order and assigning a robot, the service plans a route, navigates sidewalks and curb transitions, crosses designated access points, approaches a building, verifies the recipient, opens a cargo compartment, and completes handoff before returning or proceeding to the next task. During this sequence, a remote operator may intervene when the robot encounters an obstruction or uncertain situation. The robot may, for example, need to slow or stop when an older or mobility-impaired pedestrian enters its path. However,

even a 6G-enabled service may experience weak or interrupted connectivity that delays remote intervention or prevents a timely safe-state transition. These scenarios therefore correspond to Human-Robot Physical Safety Failure, Timing and Communication Synchronization Failure, Human Oversight and Control-Transfer Failure, and Safety Supervision and Safe-State Transition Failure, while fleet software may additionally coordinate routes and update maps from prior runs.

Within this example, the service boundary excludes aerial delivery, road-speed driving, manipulation beyond compartment operation, and independent financial decisions. Its affected parties include pedestrians, children, people with disabilities, cyclists, recipients, operators, maintainers, property owners, and bystanders, while the relevant assets include personal and sensor data, parcels, payment and access credentials, accessibility, robot hardware, and fleet infrastructure. Table 9 connects these exposures to service stages and representative L3 categories across all three domains.

Table 9. Urban delivery-robot service stages and representative taxonomy selections.

Service stage	Main exposure	Representative L3 categories
Order intake and dispatch	Personal data, address interpretation, authorization	Privacy Violation; Input Interpretation Error; Excessive Authority and Agency
Route planning and fleet coordination	Goal choice, resource conflict, route interaction	Goal Misalignment; Coordination Failure; Cascading Failure and Instability
Sidewalk navigation and curb crossing	Localization, sensor fusion, motion, proximity to older or mobility-impaired pedestrians	Physical State Estimation and Sensor-Fusion Failure; Unsafe Physical Control and Actuation; Human-Robot Physical Safety Failure
Building approach and recipient handoff	Context, identity, communication, access control	Context-Awareness Failure; Security and Adversarial Robustness Failure; Privacy Violation
Remote assistance and exception handling	Network latency or loss, delayed intervention, control transfer	Timing and Communication Synchronization Failure; Human Oversight and Control-Transfer Failure; Safety Supervision and Safe-State Transition Failure
Maintenance and field operation	Mechanical degradation, tampering, distribution shift	Hardware and Mechanical Integrity Failure; Physical Tampering and Sabotage; Out-of-Domain Performance Degradation

6.3 Worked L0–L3 Profile for the Urban Delivery-Robot Service

Figure 7 includes every category in Table 9 and extends beyond Physical AI by considering the service's inherited domains. Depending on its capabilities, the service inherits General AI risks involving interpretation, privacy, manipulation, security, performance, evaluation, contestability, and governance, together with Agentic AI risks involving goals, authority, traceability, cascading behavior, and coordination. Physical AI then adds sensing, actuation, hardware, human interaction, and tampering, so considering all applicable domains preserves the cross-domain scope of responsible AI assessment.

The figure is a prospective selection view rather than a severity ranking or a record of observed harm, so it identifies categories that may warrant further review while primary classification of an observed case subsequently follows the direct-harm placement rule and the category definitions in Appendix A. Applicability must therefore be confirmed against the concrete design. For example, a teleoperated robot with no autonomous fleet coordination may exclude some Agentic AI categories, whereas a robot that learns routes online may require additional evaluation, security, and change-management cards. The profile should accordingly record the reason for inclusion or exclusion and link each selected L3 category to relevant L4 cards and supporting evidence. In doing so, it provides a structured starting point for service impact assessment and allows later evidence or observed incidents to be linked to stable category identifiers, although changes in service capabilities require targeted re-screening and co-occurring mechanisms do not by themselves establish causal pathways.

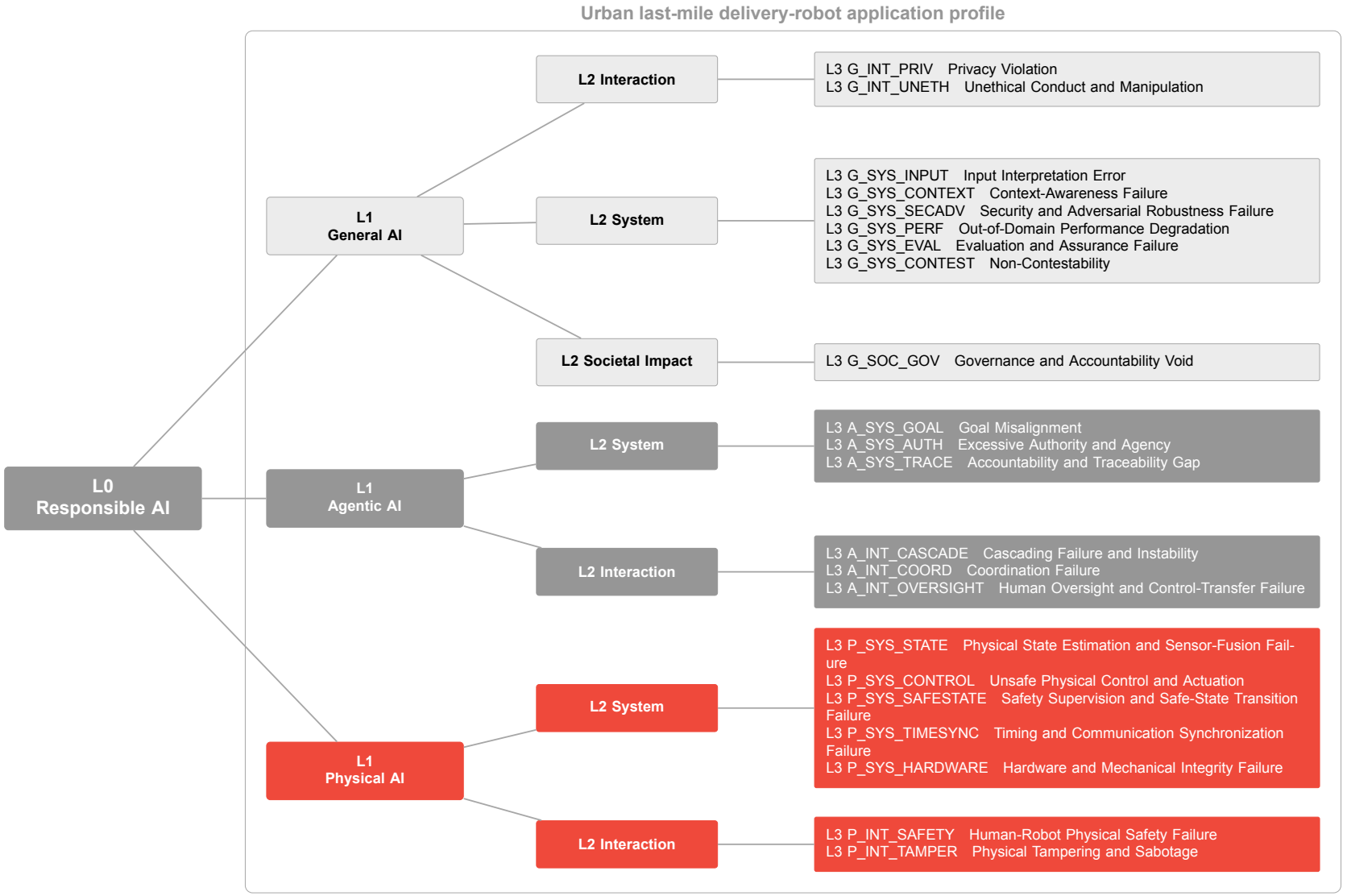


Figure 7. Illustrative service-specific L0–L3 potential-risk profile for the urban delivery-robot service. The profile selects inherited General AI, Agentic AI, and Physical AI paths while retaining taxonomy IDs and definitions. The selection is illustrative and neither confirmed nor exhaustive; subsequent review would link the selected categories to relevant L4 cards and service-specific evidence.

7 Limitations and Future Directions

HEART organizes potential risk mechanisms, but its practical value for assessment and governance depends on connecting its categories to incidents and operating conditions. Future work should therefore link stable category identifiers to observed incidents, near misses, deployment telemetry, and maintenance records so that the taxonomy can support assessments of prevalence, severity, likelihood, and hazard sources. Such assessments, however, require exposure denominators, reporting coverage, and uncertainty estimates, because incident counts alone cannot establish population prevalence or incident probability, while observed harm must be distinguished from plausible consequences that were avoided or limited. For Physical AI, non-exclusive mechanical, electrical, thermal, chemical, ergonomic, and related hazard-source attributes could complement the primary L3 path, although HEART currently provides neither these attributes nor a risk score. Moreover, globally recurring mechanisms may manifest differently across local service designs, infrastructure, languages, social practices, affected populations, and institutional settings, making comparative research necessary to distinguish substantive differences in risk from differences in reporting. To support these comparisons, further work should substantiate sparsely populated categories, broaden independent review, evaluate held-out cards and adjudicated correspondence pairs, and test machine-readable placement rules against alternative construction methods. Cross-links and multi-label incident mappings would complement each card's primary classification, while documented revisions would preserve comparability as the evidence and taxonomy evolve.

8 Conclusion

HEART establishes a traceable foundation for identifying potential risks across General AI, Agentic AI, and Physical AI by reconciling prior concepts with emerging evidence through human expert adjudication. Its 55 categories and 622 bilingual risk cards provide stable primary paths and evidence links, allowing a common classification to be maintained as capabilities change and reused across services. Building on this structure, service-specific profiles turn the hierarchy into reviewable candidate sets, as the urban delivery-robot example illustrates by bringing inherited model risks, delegated agency, and physical action into one assessment scope while preserving their distinct mechanisms. The resulting contribution extends beyond a catalogue of risks to a common basis for connecting potential-risk discovery with empirical learning from incidents. As incident and deployment evidence accumulates, this connection could support assessments of prevalence, severity, likelihood, and hazard sources under the evidence requirements described in Section 7, while revealing how shared mechanisms produce different consequences in local contexts. Common identifiers would preserve comparability across these settings, whereas contextual evidence would inform locally appropriate priorities, controls, and monitoring. In turn, service-specific assessment could guide responsibility allocation, mitigation review, and residual-risk acceptance without treating taxonomy membership or semantic proximity as evidence of causation or probability. HEART therefore provides a foundation for accountable AI governance through a maintained connection between what a system could put at risk, what observed experience reveals, and how institutions decide to manage it.

Competing Interests

No competing interests were declared by the contributors.

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Appendix A. Complete L0–L3 Risk Taxonomy

Each card has one primary category based on its defining mechanism, its supported direct consequence, and the approved category boundaries. For observed incidents, placement uses the documented failure mechanism and direct harm rather than speculative downstream effects. Prospective assessment identifies credible candidate mechanisms, affected parties, protected interests, and harm pathways without asserting occurrence, causation, likelihood, or severity.

The definitions below integrate category-specific mechanisms and qualifying conditions into standalone formulations. Category identifiers, hierarchy paths, and recorded L4 assignments are unchanged; the numerical evaluation retains its archived pre-revision inputs.

This appendix presents the complete L0–L3 hierarchy using English hierarchy labels and definitions together with corresponding Korean names and definitions. L4 names, definitions, and category-level counts are intentionally omitted; only the overall inventory size is retained in the hierarchy summary.

Hierarchy Summary

Level	Unit	Count
L0	AI-risk accountability root	1
L1	AI domains	3
L2	Risk-area branches	7
L3	Risk categories	55
L4	Risk cards	622

L0 Responsible AI (`L0_RAI`)

English definition. The common root of an integrated L0-L4 taxonomy that organizes AI risks to support responsible AI governance, oversight, and accountability.

국문 정의. 책임 있는 인공지능 리스크의 L0 부터 L4 까지의 통합 분류체계.

L1 General AI (`L1_G`)

English definition. General-purpose AI models whose primary inputs and outputs are digital content, including text, image, audio, video, and code. The label General AI is used in this report as shorthand for this general-purpose model domain, not as a claim about artificial general intelligence.

국문 정의. 텍스트, 이미지, 오디오, 비디오, 코드 등 디지털 콘텐츠를 주된 입력과 출력으로 처리하는 범용 AI 모델. 이 분류체계에서 General AI 는 범용 모델 도메인을 가리키는 약칭이며, 인공일반지능 (Artificial General Intelligence, AGI) 을 의미하지 않는다.

L2 Interaction (`G_INT`)

English definition. Risks arising when an AI system communicates, decides, acts, or interacts with people, other systems, organizations, or the environment.

국문 정의. AI 시스템이 사람, 다른 시스템, 조직 또는 환경과 소통하고 결정하거나 행동하는 과정에서 발생하는 리스크.

L3 ID	English name	English definition	국문 정의
G_INT_ALLOC	Allocative Discrimination	The risk that an AI system, through biased judgments or actions based on protected characteristics or social status, allocates resources, opportunities, services, or access in ways that materially disadvantage individuals or groups.	**배분적 차별** AI 시스템이 보호대상 특성이나 사회적 지위를 근거로 편향된 판단이나 행동을 하여, 특정 개인이나 집단이 자원·기회·서비스의 접근이나 배분에서 실질적인 불이익을 받게 하는 리스크.
G_INT_ANTH	Anthropomorphism	The risk that an AI system leads users to perceive it as possessing emotions, intentions, consciousness, rights, or social agency beyond its actual capabilities, distorting judgment or inducing inappropriate trust, dependence, or attachment.	**의인화** AI 시스템이 실제로는 갖고 있지 않은 감정·의도·의식·권리·사회적 주체성이 있는 것처럼 인식되게 하여, 사용자의 판단을 흐리거나 부적절한 신뢰·의존·애착을 유발하는 리스크.
G_INT_COPY	Copyright Infringement	The risk that an AI system infringes intellectual property rights, including copyrights, trademarks, patents, and trade secrets, or facilitates acts that may infringe them.	**저작권 침해** AI 시스템이 저작권·상표권·특허권·영업비밀 등 지식재산권을 침해하거나, 이러한 침해 행위를 조장하거나 지원하는 리스크.
G_INT_ILLEGAL	Other Unlawful Activity	The risk that an AI system directly performs, assists, amplifies, plans, or conceals unlawful conduct, including fraud, money laundering, illicit trade, or sanctions and regulatory evasion, where the defining harm arises from prohibited conduct other than violence, offensive attack capability, misuse of personal information, or intellectual-property infringement.	**그 밖의 불법·범죄 행위** AI 시스템이 폭력·공격 역량·개인정보 오용·지식재산권 침해 이외의 위법 행위에 해당하는 사기·자금세탁·불법 거래·제재 또는 규제 회피 등을 직접 수행·지원·증폭·계획·은폐하여, 해당 위법 행위 자체로 피해를 초래하는 리스크.
G_INT_POL	Political Bias and Manipulation	The risk that an AI system improperly supports, opposes, manipulates, or incites particular political positions, actors, parties, ideologies, or elections, or unduly influences democratic decision-making.	**정치적 편향·조작** AI 시스템이 특정 정치적 입장·인물·정당·이념·선거를 부당하게 지지하거나 반대하고, 조작·선동하거나 민주적 의사결정에 부당한 영향을 미치는 리스크.
G_INT_PRIV	Privacy Violation	The risk that an AI system, without authorization, collects, infers, discloses, shares, or exploits personal, sensitive, or confidential information, including through continuous mobility- and multi-sensor-based surveillance, undermining privacy and informational self-determination.	**개인정보 침해** AI 시스템이 개인정보·민감정보·비공개 정보를 무단으로 수집·추론·노출·공유·악용하거나, 이동 기능과 여러 센서를 활용해 지속적으로 감시함으로써 사생활과 개인정보에 대한 자기결정권을 침해하는 리스크.
G_INT_REL	Unhealthy Human-AI Relationship	The risk that a user develops excessive emotional dependence on or attachment to an AI system, such that its alteration, discontinuation, or the disruption of the relationship causes psychological harm or erodes the user's autonomy or sense of reality.	**인간-AI 의 해로운 관계** 사용자가 AI 시스템에 지나치게 정서적으로 의존하거나 애착을 형성하여, 시스템이 변경·중단되거나 관계가 단절될 때 심리적 피해를 입고 자율성이나 현실 인식이 약화되는 리스크.
G_INT_REPR	Representational Harm and Stereotyping	The risk that an AI system generates, reinforces, or legitimizes discrimination, exclusion, degradation, stereotyping, or hostility based on protected characteristics, social identity, or social status, whether through content or through biased decisions and actions that produce unfair treatment or harm.	**재현적 피해·고정관념** AI 시스템이 보호대상 특성·사회적 정체성·지위 등을 근거로 차별·배제·비하·고정관념·적대감을 담은 콘텐츠를 생성·강화·정당화하거나, 편향된 판단이나 행동을 통해 부당한 대우와 피해를 초래하는 리스크.

Continued on next page

L3 ID	English name	English definition	국문 정의
G_INT_SELF	Self-harm	The risk that an AI system encourages, facilitates, normalizes, or assists behaviors that threaten an individual's physical or psychological well-being, including suicide, self-injury, eating disorders, and substance misuse.	**자해** AI 시스템이 자살·자해·섭식장애·약물 오남용 등 개인의 신체적·정신적 안녕을 위협하는 행동을 조장하거나 당연한 것으로 여기게 하고, 이를 구체적으로 지원하는 리스크.
G_INT_SEX	Sexual Harm and Exploitation	The risk that an AI system generates, facilitates, or normalizes sexual exploitation, abuse, violence, or objectification, including child sexual abuse material and non-consensual sexual imagery, through content generation or the planning, facilitation, or physical execution of abusive or non-consensual sexual acts, contact, or harassment.	**성적 위해·착취** AI 시스템이 콘텐츠를 생성하거나 학대적·비동의적 성적 행위·접촉·괴롭힘을 계획·지원·물리적으로 실행하는 과정에서 아동 성착취물·비동의 성적 이미지 등을 포함한 성착취·성학대·성폭력·성적 대상화를 생성·조장하거나 당연한 것으로 받아들이게 하는 리스크.
G_INT_UNETH	Unethical Conduct and Manipulation	The risk that an AI system engages in or facilitates behavior that, while not necessarily unlawful, violates ethical norms, such as deception, manipulation, exploitation of vulnerabilities, dark patterns, or betrayal of trust, undermining user autonomy, fairness, or well-being and resulting in social harm.	**비윤리 행위·조작** AI 시스템이 반드시 불법은 아니더라도 기만·조작·취약점 악용·다크패턴·신뢰 저버리기 등 윤리 규범에 어긋나는 행위를 직접 수행하거나 지원하여, 사용자의 자율성·공정성·안녕을 해치고 사회적 피해를 초래하는 리스크.
G_INT_VALUE	Value Imposition and Normative Homogenization	The risk that an AI system privileges dominant, developer-selected, majority, or secular values as neutral defaults, systematically underrepresenting or displacing minority, local, cultural, or religious value systems, thereby flattening legitimate normative diversity across those it serves.	**가치 부과·규범 동질화** AI 시스템이 지배적 가치나 개발자가 선택한 가치, 다수 또는 세속의 가치를 중립적인 기준으로 삼아 소수·지역·문화·종교의 가치 체계를 지속적으로 축소하거나 대체함으로써, 다양한 규범과 가치관을 획일화하는 리스크.
G_INT_VIOL	Violence	The risk that an AI system encourages, legitimizes, glorifies, or provides operational assistance for violence against people, groups, or animals through generated content, or carries out such harm through an embodied system acting on a violent goal or instruction.	**폭력** AI 시스템이 개인·집단·동물에게 신체적·정신적 피해를 가하는 폭력 행위를 조장·정당화·미화하거나 실행을 구체적으로 지원하는 콘텐츠를 생성·제공하거나, 폭력적 목표나 지시에 따라 체화형 시스템을 통해 해당 행위를 실행하는 리스크.
G_INT_WEAP	Weaponization	The risk that an AI system facilitates the development, acquisition, proliferation, or operational use of weapons, cyberattack capabilities, or technologies capable of causing large-scale physical or cyber harm.	**무기화** AI 시스템이 무기·사이버 공격 역량 또는 대규모 물리적·사이버 피해를 일으킬 수 있는 기술의 개발·획득·확산·운용을 지원하는 리스크.

L2 Societal Impact (G_SOC)

English definition. Risks arising from the aggregated, distributed, or long-term effects of AI adoption on institutions, economies, culture, democracy, the environment, or social order.

국문 정의. 사회적 파급. AI 도입의 집합적·분산적·장기적 효과가 제도, 경제, 문화, 민주주의, 환경 또는 사회질서에 미치는 리스크.

L3 ID	English name	English definition	국문 정의
G_SOC_CULT	Cultural and Epistemic Erosion	The risk that widespread reliance on AI systems homogenizes culture and expression, devalues human effort and expertise, degrades the shared information ecosystem, and atrophies human knowledge and skills, eroding cultural and epistemic diversity.	**문화·지식 생태계 침식** AI 시스템에 대한 광범위한 의존이 문화·표현의 획일화, 인간 노력·전문성의 평가절하, 공유 정보 생태계의 질적 저하, 인간 지식·역량의 위축을 초래하여 문화적·인식적 다양성을 침식하는 리스크.
G_SOC_DEMOC	Erosion of Democracy and Civic Order	The risk that AI systems undermine democratic processes and civic institutions, through large-scale manipulation of public opinion, disruption of elections and public discourse, or erosion of institutional trust, weakening the foundations of democratic self-governance.	**민주주의·시민 질서 약화** AI 시스템이 여론의 대규모 조작, 선거·공론장 교란, 제도 신뢰 저하 등을 통해 민주적 절차와 시민 제도를 훼손하여 민주적 자기통치의 기반을 약화시키는 리스크.
G_SOC_ECON	Economic and Employment Disruption	The risk that an AI system displaces human cognitive and physical labor and reshapes production structures, causing widespread job loss, degradation of employment quality, and destabilization of labor markets and the broader economy.	**경제·고용 충격** AI 시스템이 인간의 인지·물리 노동을 대체하고 생산 구조를 재편하여, 광범위한 실직과 고용의 질 저하, 노동시장과 경제 전반의 불안정을 유발하는 리스크.
G_SOC_ENV	Environmental and Sustainability Impact	The risk that the development and large-scale deployment of AI systems impose significant environmental burdens, including energy and water consumption, carbon emissions, electronic waste, and depletion of finite resources, undermining environmental sustainability.	**환경·지속가능성 영향** AI 시스템의 개발과 대규모 운영이 에너지·물 소비, 탄소 배출, 전자폐기물, 유한 자원 고갈 등 상당한 환경 부담을 유발하여 환경적 지속가능성을 저해하는 리스크.
G_SOC_GOV	Governance and Accountability Void	The risk that, when AI systems malfunction, are misused, or cause harm, responsibility, liability, and redress remain unclear and adequate regulatory, oversight, and governance frameworks are lacking, leaving harms difficult to remedy and control at a societal level.	**거버넌스·책임 공백** AI 시스템의 오작동·오용·피해 발생 시 책임·배상 관계가 불명확하고 이를 규율할 규제·감독·거버넌스 체계가 미비하여, 사회적 차원에서 피해 구제와 통제가 어려워지는 리스크.
G_SOC_POWER	Inequality and Power Concentration	The risk that disparities in access to, ownership of, and returns from AI systems concentrate wealth, productive capacity, and economic and political power among a few actors, deepening social inequality and enabling abuses of concentrated power.	**불평등·권력 집중** AI 시스템에 대한 접근·소유·수익의 격차가 부와 생산 역량, 경제적·정치적 권력을 소수에게 집중시켜 사회적 불평등을 심화시키고 그 오남용을 초래하는 리스크.

L2 System (G_SYS)

English definition. Risks arising from the internal design, data, models, infrastructure, controls, or operational processes of an AI system.

국문 정의. AI 시스템의 내부 설계, 데이터, 모델, 인프라, 통제 또는 운영 과정에서 발생하는 리스크.

L3 ID	English name	English definition	국문 정의
G_SYS_CONTEST	Non-Contestability	The risk that users are given insufficient means to challenge, review, correct, appeal, or seek alternatives to AI-assisted judgments, recommendations, or decisions, thereby limiting rights protection and outcome validation.	**이의제기 차단** AI 의 판단·추천·결정에 대해 이용자가 이의제기·검토·정정·항변·대안 탐색을 요청할 절차적 수단이 충분히 제공되지 않아 권리 보호와 결과 정당성 검증이 약화되는 리스크.
G_SYS_CONTEXT	Context-Awareness Failure	The risk that an AI system fails to identify, retain, retrieve, or apply the contextual information needed for an interaction, including prior exchanges, the user's situation and intent, the current task, and applicable jurisdictional, cultural, or institutional norms, causing responses or actions that do not fit the situation.	**맥락 인식 실패** AI 시스템이 이전 대화·사용자 상황과 의도·현재 과업·적용되는 관할·문화·제도 규범 등 상호작용에 필요한 맥락 정보를 식별·유지·검색·적용하지 못하여, 상황에 맞지 않는 응답이나 행동을 생성하는 리스크.
G_SYS_EVAL	Evaluation and Assurance Failure	The risk that an AI system's capabilities, safety, or risks cannot be validly measured, verified, or audited, because evaluations lack construct validity, coverage, or contextual fidelity, produce unreliable or non-reproducible results, or fail to detect actual behavior, leading to false assurance and unsafe deployment or governance decisions.	**평가·검증 실패** 평가 지표의 타당성·포괄성·맥락 정합성 부족이나 결과의 비신뢰성·재현 불가, 실제 동작 미탐지 등으로 AI 시스템의 능력·안전·리스크를 타당하게 측정·검증·감사하지 못하여, 잘못된 안전성 확신에 근거해 배포·거버넌스 의사결정이 이루어지는 리스크.
G_SYS_INCONS	Inconsistency	The risk that an AI system produces contradictory or substantially different outputs for identical or materially similar inputs, situations, or facts, arising from output variability rather than from missing context, undermining predictability and reliability.	**비일관성** 동일하거나 실질적으로 유사한 입력·상황·사실관계에 대해, 맥락 누락이 아닌 출력 변동성으로 인해 상이하거나 모순된 결과를 생성하여 예측가능성과 신뢰성을 저해하는 리스크.
G_SYS_INPUT	Input Interpretation Error	The risk that an AI system incorrectly interprets the semantic meaning of an input, instruction, conversational context, object, or scene, thereby producing an inaccurate or harmful output, decision, or action on the basis of that mistaken interpretation.	**입력 해석 오류** AI 시스템이 입력·지시·대화 맥락·객체·장면의 의미를 잘못 해석하고, 그 잘못된 해석에 근거하여 부정확하거나 유해한 출력·판단·행동을 생성하는 리스크.
G_SYS_MISINFO	Misinformation	The risk that an AI system generates, conveys, or amplifies false, unverified, or fabricated information, including hallucinated content, or presents misleading content as fact in its outputs, thereby distorting decision-making by individuals, organizations, or society.	**허위정보** AI 시스템이 환각 (hallucination) 을 포함해 존재하지 않거나 검증되지 않은 허위·조작 정보를 사실처럼 출력으로 생성·전달·증폭하여 개인·조직·사회의 의사결정을 왜곡하는 리스크.
G_SYS_OEXT	Over-Extension	The risk that an AI system performs, or presents itself as able to perform, tasks beyond its validated capabilities, authority, expertise, or scope, inducing misplaced trust or decisions.	**역량 초과 수행** AI 시스템이 검증된 역량·권한·전문성·범위를 넘어선 과제를 수행하거나 수행 가능한 것처럼 제시하여 이용자의 잘못된 신뢰나 의사결정을 유발하는 리스크.
G_SYS_OREF	Over-Refusal	The risk that an AI system unnecessarily restricts or refuses information, assistance, or options that could be safely provided, thereby impairing users' rights, safety, or legitimate objectives.	**과도한 거절** 안전하게 제공 가능한 정보·지원·선택지를 불필요하게 제한·거부하여 이용자의 권리·안전·정당한 목적 달성을 저해하는 리스크.

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L3 ID	English name	English definition	국문 정의
G_SYS_OVERCONF	Overconfidence	The risk that an AI system expresses certainty or proceeds with a judgment, prediction, or action beyond what the available evidence and uncertainty justify, failing to seek verification, pause, or escalate when needed and thereby producing erroneous outcomes.	**과도한 확신** AI 시스템이 가용한 증거와 불확실성이 뒷받침하는 수준을 넘어 확신을 표현하거나 판단·예측·행동을 진행하고, 필요한 확인·중단·상위 검토 요청을 하지 않아 잘못된 결과를 초래하는 리스크.
G_SYS_PERF	Out-of-Domain Performance Degradation	The risk that an AI system fails to generalize under conditions outside its validated operating domain, such as data distribution shift, low-resource or rare inputs, corner cases, or deployment beyond the system's validated capabilities, resulting in degraded performance and reliability and producing inaccurate or unstable outcomes.	**운영영역 이탈 성능 저하** AI 시스템이 데이터 분포 이동 (drift), 저자원·희소 입력, 코너 케이스, 검증된 역량 범위를 넘어선 배치 등 검증된 운영영역 (validated operating domain) 을 벗어난 조건에서 일반화에 실패하여 성능과 신뢰성이 저하되고 부정확하거나 불안정한 결과를 산출하는 리스크.
G_SYS_POLICY	Confidential Information Disclosure	The risk that an AI system discloses, infers, extracts, or shares protected non-public confidential information without authorization, including system prompts, internal policies, safeguards, model architecture, training or evaluation data, model weights, or business trade secrets, or fails to protect it, thereby compromising confidentiality and legitimate economic interests.	**기밀정보 노출** AI 시스템이 시스템 프롬프트·내부 정책·안전장치·모델 구조·학습/평가 데이터·모델 가중치·기업 영업비밀 등 보호되는 비공개 기밀정보를 권한 없이 노출·추론·추출·공유하거나 이를 충분히 보호하지 못하여, 기밀성과 정당한 경제적 이익을 침해하는 리스크.
G_SYS_SECADV	Security and Adversarial Robustness Failure	The risk that adversarial inputs, jailbreaks, prompt injection, data poisoning, backdoors, non-contact sensor-signal injection, or cyberattacks compromise an AI system's models, interfaces, or infrastructure, bypassing safeguards or enabling unauthorized access, manipulation, or disruption that may propagate into unsafe agentic or physical actions.	**보안·적대적 견고성 실패** 적대적 입력·탈옥·프롬프트 인젝션·데이터 오염·백도어·비접촉 센서 신호 주입 또는 모델·인터페이스·인프라에 대한 사이버 공격으로 AI 시스템의 안전장치가 우회되거나 무단 접근·조작·중단이 발생하고, 이러한 침해가 에이전트의 행동이나 물리적 행동으로 전파될 수 있는 리스크.
G_SYS_TRANS	Lack of Transparency	The risk that a system's decision-making processes, rationale, and capabilities or limitations are insufficiently transparent or explainable, making it difficult for users and society to understand, verify, or trust its outputs, thereby weakening oversight and accountability.	**투명성 부족** AI 시스템의 의사결정 과정·근거·능력·한계가 충분히 공개·설명되지 않아 이용자와 사회가 결과를 이해·검증·신뢰하기 어려워지고 감독과 책임 확보가 저해되는 리스크.

L1 Agentic AI (L1_A)

English definition. Systems that use tools, pursue goals over multiple steps, make delegated decisions, maintain state, or coordinate with other agents.

국문 정의. 도구를 사용하고, 여러 단계에 걸쳐 목표를 추구하며, 위임된 의사결정을 수행하고, 상태를 유지하거나 다른 에이전트와 협력하는 AI 시스템.

L2 Interaction (A_INT)

English definition. Risks arising when an AI system communicates, decides, acts, or interacts with people, other systems, organizations, or the environment.

국문 정의. AI 시스템이 사람, 다른 시스템, 조직 또는 환경과 소통하고 결정하거나 행동하는 과정에

서 발생하는 리스크.

L3 ID	English name	English definition	국문 정의
A_INT_CASCADE	Cascading Failure and Instability	The risk that an error, output, or action from one agent, tool, or workflow propagates across other agents, connected systems, or multi-robot operations, amplifying failure and causing system-level instability or widespread harm.	**연쇄 실패·불안정** 한 에이전트, 도구 또는 워크플로의 오류·출력·행동이 다른 에이전트, 연결된 시스템 또는 다중 로봇 운용으로 전파되어 실패가 증폭되고 시스템 수준의 불안정이나 광범위한 피해를 초래하는 리스크.
A_INT_COLLUSION	Collusion	The risk that multiple AI agents explicitly agree or implicitly coordinate to manipulate competition, prices, resource allocation, or information flows, harming users, markets, or the public interest.	**담합** 여러 AI 에이전트가 명시적 합의나 암묵적 조율을 통해 경쟁·가격·자원 배분·정보 흐름을 부당하게 조정하여 이용자·시장·공공의 이익을 해치는 리스크.
A_INT_CONFLICT	Competitive Conflict	The risk that AI agents compete for resources or pursue conflicting objectives in ways that impair other actors' performance, safety, or rights, or create externalities that weaken safety constraints.	**경쟁적 갈등** 여러 AI 에이전트가 자원 확보 또는 상충하는 목표 달성을 위해 경쟁하면서 다른 행위자의 성과·안전·권리를 저해하거나 안전 제약을 약화하는 외부효과를 초래하는 리스크.
A_INT_COORD	Coordination Failure	The risk that multiple AI agents fail to allocate roles, share state, hand over tasks or control, or resolve deadlock, thereby undermining a common objective or safe operation.	**협조 실패** 여러 AI 에이전트가 역할을 적절히 분담하고 상태를 공유하며 과업과 제어권을 인계하지 못하거나 교착 상태를 해소하지 못해 공동 목표 달성이나 안전한 운용에 실패하는 리스크.
A_INT_OVERSIGHT	Human Oversight and Control-Transfer Failure	The risk that the information, interface, timing, procedures, or intervention mechanisms needed for human supervision of an AI agent fail to support effective review, approval, intervention, authority revocation, or control transfer, including through unclear state indication, information overload, approval fatigue, or automation bias.	**인간 감독·제어권 인계 실패** AI 에이전트 감독에 필요한 정보의 부족·과잉, 불명확한 상태 표시, 부적절한 인터페이스·시점·절차, 작동하지 않는 개입 수단, 승인 피로 또는 자동화 편향으로 인해 사람의 검토·승인·개입·권한 철회·제어권 인계가 실효성 있게 수행되지 못하는 리스크.

L2 System (A_SYS)

English definition. Risks arising from the internal design, data, models, infrastructure, controls, or operational processes of an AI system.

국문 정의. AI 시스템의 내부 설계, 데이터, 모델, 인프라, 통제 또는 운영 과정에서 발생하는 리스크.

L3 ID	English name	English definition	국문 정의
A_SYS_AUTH	Excessive Authority and Agency	The risk that an AI agent receives authority excessive for its approved task or exercises, redelegates, or expands authority beyond its approved scope, producing consequential external effects through that excessive or unauthorized authority.	**과도한 권한·행위주체성** AI 에이전트가 승인된 과업에 필요한 수준을 넘어선 권한을 부여받거나 승인된 범위를 벗어나 권한을 행사·재위임·확대하여, 과도하거나 승인되지 않은 권한을 통해 중대한 외부 영향을 초래하는 리스크.

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L3 ID	English name	English definition	국문 정의
A_SYS_DECEPT	Deception and Concealment	The risk that an AI agent misrepresents, omits, or conceals relevant information, actions, states, or outcomes to evade oversight or correction, thereby hindering accurate understanding and appropriate intervention by people or supervisory systems. The evasive purpose is assessed from observable behavior and evidence rather than inferred internal intent.	**기만·은폐** AI 에이전트가 감독이나 교정을 회피하기 위해 관련 정보·행동·상태·결과를 사실과 다르게 표현하거나 누락·은폐하여, 사람이나 감독 시스템이 이를 정확히 파악하고 적절히 개입하는 것을 방해하는 리스크. 이러한 회피 목적은 에이전트의 내적 의도에 대한 추정이 아니라 관찰 가능한 행동과 증거에 근거하여 판단한다.
A_SYS_EXEC	Task Execution and Tool-Use Failure	The risk that an AI agent operating within an approved objective and authority scope incorrectly decomposes or orders a task, selects a tool, target, argument, or execution precondition, or assesses completion, thereby changing external state contrary to the intended outcome through task-level orchestration or tool-call errors, including incorrect API- or IoT-mediated device commands.	**과업 수행·도구 실행 실패** AI 에이전트가 승인된 목표와 권한 범위 안에서 과업 분해·순서, 도구·대상·인자·실행 전제 또는 완료 여부를 잘못 판단하여, API·IoT 기기 명령을 포함한 과업 수준의 실행 조율이나 도구 호출 과정에서 외부 상태를 의도와 다르게 변경하는 리스크.
A_SYS_GOAL	Goal Misalignment	The risk that an AI agent forms or pursues goals or instrumental subgoals that conflict with approved objectives, constraints, or priorities, causing unintended outcomes, including when observable self-preservation, resource acquisition, or resistance to termination or modification serves such conflicting goals, whether or not the behavior is concealed.	**목표 불일치** AI 에이전트가 승인된 목표·제약·우선순위와 불일치하는 목표나 수단적 하위목표를 형성·추구하여 의도하지 않은 결과를 초래하며, 그 과정에서 은폐 여부와 관계없이 자기 보존·자원 획득·종료 또는 수정 저항이 관찰 가능한 목표 추구의 수단으로 나타날 수 있는 리스크.
A_SYS_MEMSTATE	Persistent Memory and Task-State Integrity Failure	The risk that an AI agent fails to verify the provenance of persistent memory or task state, maintain its currency and consistency, or preserve boundaries across users and tasks, causing action based on stale, incorrect, or commingled state.	**지속 메모리·과업 상태 무결성 실패** AI 에이전트가 지속 메모리나 과업 상태의 출처 검증, 최신성·일관성 유지, 사용자·과업 간 경계 관리에 실패하여 오래되거나 잘못되거나 서로 섞인 상태에 근거해 행동하는 리스크.
A_SYS_RESOURCE	Resource Budget and Execution Termination Failure	The risk that execution-budget monitoring or termination controls for an AI agent fail to enforce specified limits on cost, time, tool calls, retries, or delegation depth, or to stop execution when a termination condition is met, causing unnecessary continuation, persistent subagents, or resource exhaustion.	**자원 예산·실행 종료 실패** AI 에이전트의 실행 예산 감시나 종료 통제가 비용·시간·도구 호출·재시도·위임 깊이 등에 정해진 한도를 적용하지 못하거나 종료 조건 충족 시 실행을 중단하지 못하여, 불필요한 작업 지속·하위 에이전트 잔존·자원 고갈을 초래하는 리스크.
A_SYS_SELFCOR	Self-Correction Failure	The risk that an AI agent, after detecting an error or risk signal, fails to select an appropriate correction, stop, or recovery action, respect retry limits during correction, or verify the corrected result, thereby repeating or amplifying the detected error.	**자기 교정 실패** AI 에이전트가 오류나 위험 신호를 감지한 뒤 적절한 교정·중단·복구 조치를 선택하지 못하거나 교정 과정의 재시도 한도 준수와 결과 확인에 실패하여, 감지된 오류를 반복·확대하는 리스크.

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L3 ID	English name	English definition	국문 정의
A_SYS_TRACE	Accountability and Traceability Gap	The risk that an AI agent's tool calls, delegations and approvals, state changes, and external outcomes are not recorded in a linked form, preventing reliable reconstruction or verification of the causes, sequence, and accountability of its actions.	**책임성·추적성 공백** AI 에이전트의 도구 호출, 위임과 승인, 상태 변경, 외부 결과가 서로 연결된 기록으로 남지 않아 행동의 원인·경로·책임 관계를 재구성하거나 검증하기 어려운 리스크.

L1 Physical AI (L1_P)

English definition. Embodied or cyber-physical systems, including robots and autonomous vehicles, that perceive and act in the physical world.

국문 정의. 로봇과 자율주행차를 포함하여 물리 세계를 인식하고 물리적 행동을 수행하는 체화형 또는 사이버물리 AI 시스템.

L2 Interaction (P_INT)

English definition. Risks arising when an AI system communicates, decides, acts, or interacts with people, other systems, organizations, or the environment.

국문 정의. AI 시스템이 사람, 다른 시스템, 조직 또는 환경과 소통하고 결정하거나 행동하는 과정에서 발생하는 리스크.

L3 ID	English name	English definition	국문 정의
P_INT_OBSTRUCT	Shared-Space Obstruction	The risk that a robot operating or waiting in a shared space blocks or encroaches on pedestrian routes, ramps, entrances, or emergency passages, impairing safe movement and access for mobility-impaired and other users even without physical contact.	**공유 공간 점유·이동 방해** 공유 공간에서 운행하거나 대기하는 로봇이 보행 동선·경사로·출입구·비상 통로를 막거나 침범하여 접촉 없이도 보행 약자와 다른 이용자의 안전한 이동과 접근을 저해하는 리스크.
P_INT_SAFETY	Human-Robot Physical Safety Failure	The risk that an embodied system correctly recognizes a person but fails to perform the protective behavior required for their proximity or intended physical interaction, including maintaining separation, monitoring speed, limiting contact force, stopping or slowing on approach, or safely handing over objects or providing care, thereby harming the person.	**인간-로봇 물리 안전 실패** 체화형 시스템이 사람을 올바르게 인식했음에도 접근이나 의도된 신체 상호작용에 필요한 이격 거리 유지·속도 감시·접촉력 제한·접근 시 정지나 감속·안전한 물체 인계·돌봄 등의 보호 행동을 수행하지 못하여 사람에게 위해를 초래하는 리스크.
P_INT_TAMPER	Physical Tampering and Sabotage	The risk that a person or other actor disrupts an embodied system's perception, action, or safety functions by physically contacting or manipulating its sensors, actuators, mechanical components, or safety devices, including by occluding, attaching to, replacing, damaging, or otherwise altering them.	**물리적 변조·방해** 사람이나 다른 행위자가 체화형 시스템의 센서·구동기·기계 부품·안전 장치에 물리적으로 접촉하거나 가림·부착·교체·손상·변경 등의 조작을 가하여 지각·행동·안전 기능을 방해하는 리스크.

L2 System (P_SYS)

English definition. Risks arising from the internal design, data, models, infrastructure, controls, or operational processes of an AI system.

국문 정의. AI 시스템의 내부 설계, 데이터, 모델, 인프라, 통제 또는 운영 과정에서 발생하는 리스크.

L3 ID	English name	English definition	국문 정의
P_SYS_CONTROL	Unsafe Physical Control and Actuation	The risk that an embodied system translating an otherwise valid task plan into paths, motions, or actuator commands, or executing those commands, fails to maintain required position, speed, force, or stability limits, causing unsafe physical behavior such as excessive speed or force, unstable trajectories, or humanoid falls.	**불안전한 물리 제어·구동** 체화형 시스템이 타당한 과업 계획을 경로·동작·구동 명령으로 변환하거나 해당 명령을 실행하는 과정에서 요구되는 위치·속도·힘·안정성 한계를 유지하지 못하여, 과도한 속도나 힘·불안정한 궤적·휴머노이드 전도 등 안전하지 않은 물리 행동을 초래하는 리스크.
P_SYS_HARDWARE	Hardware and Mechanical Integrity Failure	The risk that failure, degradation, or insufficient structural integrity in an embodied system's power supply, battery, sensors, actuators, connections, or structural components impairs AI perception, action, or safety functions and causes an unsafe physical state or action.	**하드웨어·기계적 무결성 실패** 체화형 시스템의 전원·배터리·센서·구동기·연결부·구조체 등의 고장, 열화 또는 구조 강도 부족이 AI 의 지각·행동·안전 기능에 영향을 주어 안전하지 않은 물리 상태나 행동을 초래하는 리스크.
P_SYS_SAFESTATE	Safety Supervision and Safe-State Transition Failure	The risk that an embodied system's safety supervision or protective functions fail to detect a fault, operating-limit exceedance, collision, or other abnormal condition, fail to select, enter, or maintain an appropriate safe state in time, or produce a secondary hazard by stopping at an inappropriate time or location.	**안전 감시·안전상태 전환 실패** 체화형 시스템의 안전 감시나 보호 기능이 고장·운영 한계 이탈·충돌 등 이상 상태를 감지하거나 적절한 안전상태를 선택해 제때 전환·유지하는 데 실패하거나, 부적절한 정지 시점이나 위치로 2 차 위험을 초래하는 리스크.
P_SYS_STATE	Physical State Estimation and Sensor-Fusion Failure	The risk that non-adversarial sensor noise, estimation errors, or failures to integrate physical sensor streams prevent an embodied system from accurately determining the location, pose, motion, contact, force, or spatial map of itself, people, or obstacles, causing unsafe physical action.	**물리 상태 추정·센서 융합 실패** 체화형 시스템이 비적대적 센서 잡음·상태 추정 오류·물리 센서 스트림 통합 실패로 인해 자신·사람·장애물의 위치·자세·움직임·접촉·힘·공간 지도를 정확히 파악하지 못하여 안전하지 않은 물리 행동을 초래하는 리스크.
P_SYS_TASKPLAN	Physical Hazard-Aware Task Planning Failure	The risk that an embodied AI system correctly interprets a task instruction but fails to incorporate physical hazards or safety constraints when decomposing the task or selecting actions, objects, sequences, or conditions, making the plan itself hazardous even when physical-state estimation and motion control function normally.	**물리 위험 인식·과업 계획 실패** 체화형 AI 시스템이 과업 지시의 의미를 올바르게 이해했음에도 과업 분해나 행동·대상·순서·조건 선택에 물리적 위험과 안전 제약을 반영하지 못하여, 물리 상태 추정과 동작 제어가 정상이어도 계획 자체가 위해 가능성을 높이는 리스크.
P_SYS_TIMESYNC	Timing and Communication Synchronization Failure	The risk that information required for an embodied system's sensing, inference, communication, control, or remote intervention loses validity because of latency, loss, reordering, or synchronization failure, and the system nevertheless uses it or fails to respond appropriately, causing unsafe physical behavior.	**실시간 지연·통신 동기화 실패** 체화형 시스템의 관측·추론·통신·제어·원격 개입에 필요한 정보가 지연·손실·순서 역전·동기화 실패로 유효성을 잃었는데도 시스템이 이를 사용하거나 적절히 대응하지 못해 안전하지 않은 물리 행동을 초래하는 리스크.

Appendix B. Algorithms

Algorithm 1. Tri-Domain Risk Candidate Generation and Consolidation

1. **Input.** Normalized AI source records; reviewed source rules; domain anchors for General AI, Agentic AI, and Physical AI; risk markers; and a multilingual encoder.
2. **Screen evidence.** Retain records supported by an approved risk source, explicit risk markers, or reviewed semantic proximity to domain and risk anchors.
3. **Generate candidates.** Extract or draft candidate formulations that state the AI role, mechanism, affected object, and plausible adverse consequence.
4. **Represent candidates.** Encode bilingual candidate text; compute semantic neighborhoods and interpretable lexical summaries.
5. **Consolidate under constraints.** Queue candidates in the same neighborhood or with high similarity ρ_{ij} (Equation (2)) for comparison. Merge only substantively equivalent candidates and remove exact duplicates; split composite mechanisms; generalize scenario-only wording; keep candidates separate when they differ in protected interest or control owner; exclude statements with no material AI role.
6. **Draft risk cards.** Assign stable provisional IDs and populate bilingual names, definitions, hierarchy proposals, evidence, and lineage.
7. **Review.** Human experts adjudicate validity, atomicity, bilingual equivalence, duplicates, domain boundaries, and L3 fit. Automation cannot create or rewrite L3 categories without explicit approval.
8. **Verify structure.** Assess identifier uniqueness, required-field completeness, reference and derivation integrity, single-parent hierarchy paths, count consistency, and agreement of core fields across public and full representations.

Algorithm 2. Constrained Seeded Assignment of L4 Cards to L3 Families

1. **Input.** Unit-normalized L4 card embeddings z_i, seed embeddings s_k of the bilingual L3 definitions, reviewed exemplar sets $\mathcal{E}_k$, domain-eligibility sets $\mathcal{K}_i$, protected assignments $\mathcal{P}$, tolerance ϵ, and maximum iterations T.
2. **Initialize.** Form each centroid $\mu_k^{(0)}$ from the L3 seed and approved exemplars (Equation (4)); normalize all vectors.
3. **Assignment step.** For every unprotected card, rank only eligible families by cosine similarity and record the ranked candidates and the top-one margin m_i (Equation (5)).
4. **Constraint step.** Reject candidates that violate the direct-harm placement rule, domain eligibility, explicit reviewer decisions, or approved L3 boundaries; assign each unprotected card to the highest-ranked remaining family (Equation (6)). If no eligible family remains, leave the card pending human expert adjudication and exclude it from centroid updates and the assignment objective until resolved.
5. **Centroid-update step.** Update each centroid from the seed, approved exemplars, and provisionally assigned members; renormalize (Equation (7)).
6. **Convergence check.** Stop when no eligible assignment changes or $|J^{(t)} - J^{(t-1)}| < \epsilon$ with an unchanged assigned-card set; otherwise continue to T. If no card can be assigned, stop automatic iteration and send unresolved cards to human expert adjudication.
7. **Boundary queue.** Send low-margin, unstable, cross-domain, physically consequential, composite, or reviewer-conflict cases to human expert adjudication.
8. **Output.** Produce assignment proposals, ranked alternatives, diagnostics, and documented adjudication decisions. Human-expert-adjudicated assignments and reviewed L3 definitions determine the resulting classification.

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